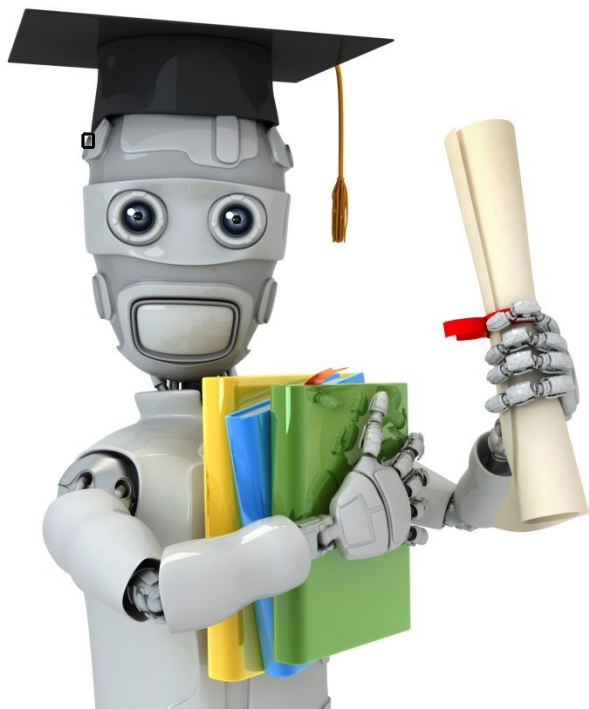




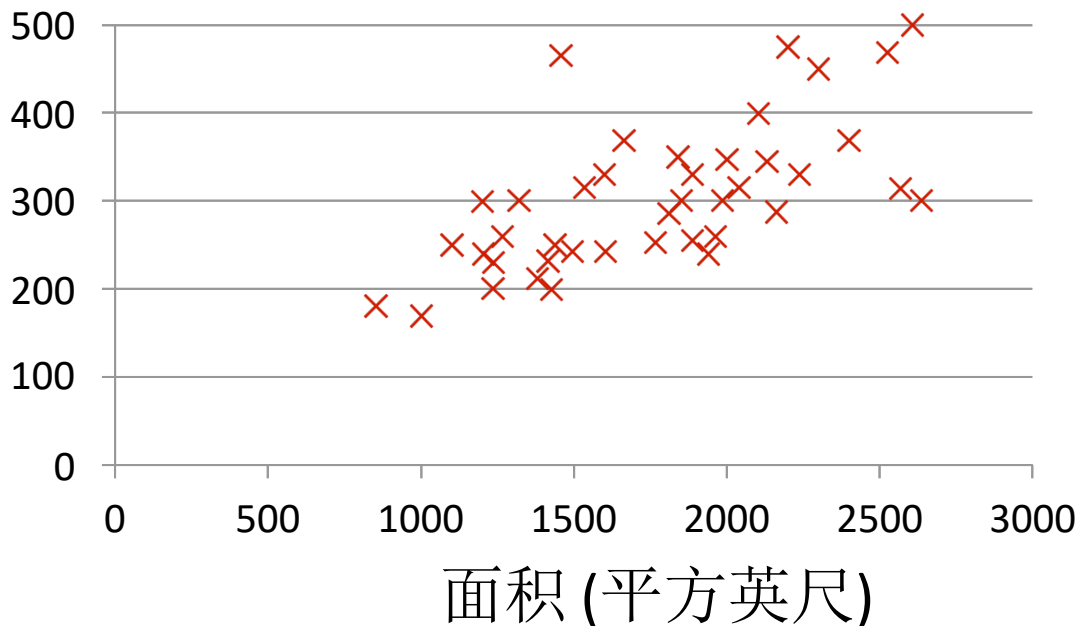
机器学习

单变量线性回归

模型表示

房屋价格 (波特兰)

价格 (每
1000刀为单
位)



监督学习

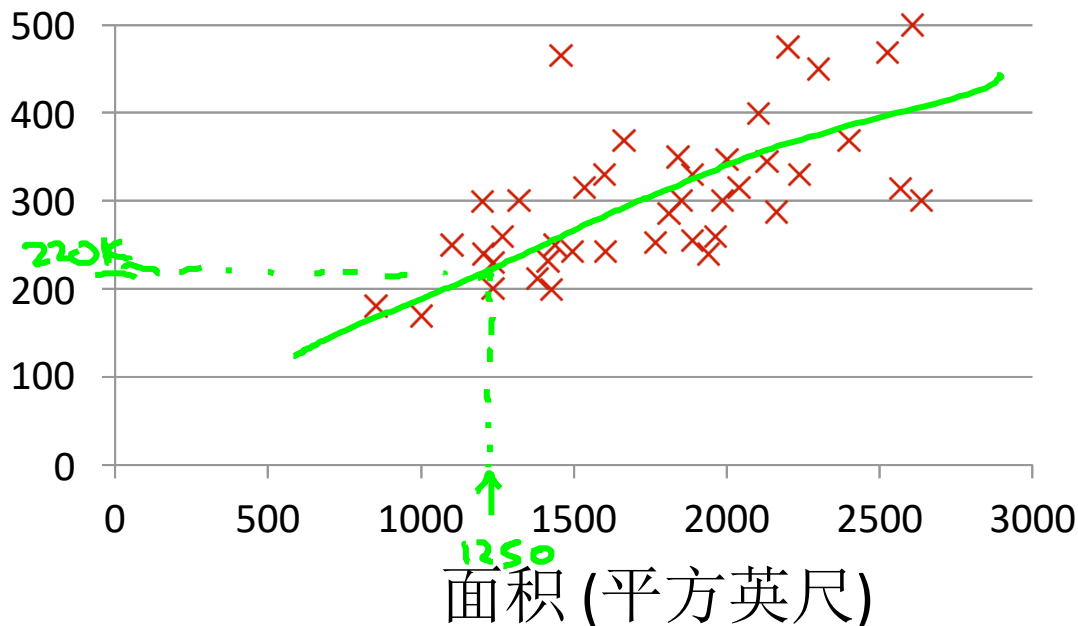
在数据中得每一个例子都给出“正确答案”。

回归问题

预测正确的价格输出

房屋价格 (波特兰)

价格 (每
1000刀为单
位)



监督学习

在数据中得每一个例子都给出“正确答案”。

回归问题

预测正确的价格输出

房屋价格 训练集(波特兰)

Size in feet ² (x)	价格 (\$) in 1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

} $m = 47$

注释:

m = 训练个数

x 's = “输入” 变量 / 特征

y 's = “输出” 变量 / “目标” 变量

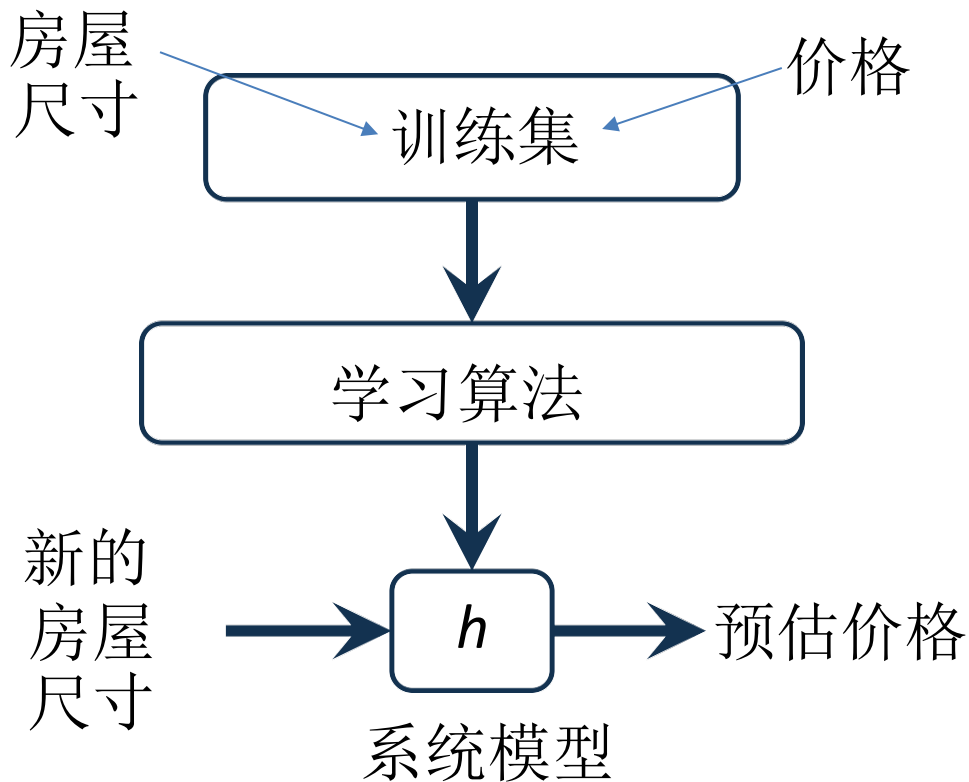
(x, y) - one training example

$(x^{(i)}, y^{(i)})$ - i th training example

$$x^{(1)} = 2104$$

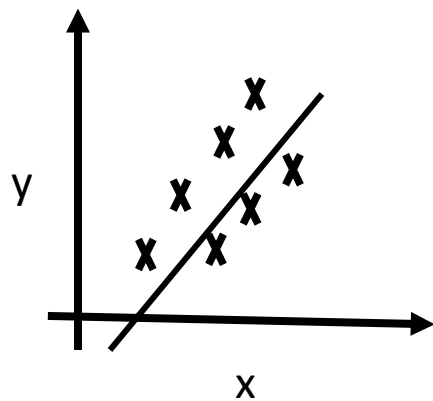
$$x^{(2)} = 1416$$

$$y^{(1)} = 460$$

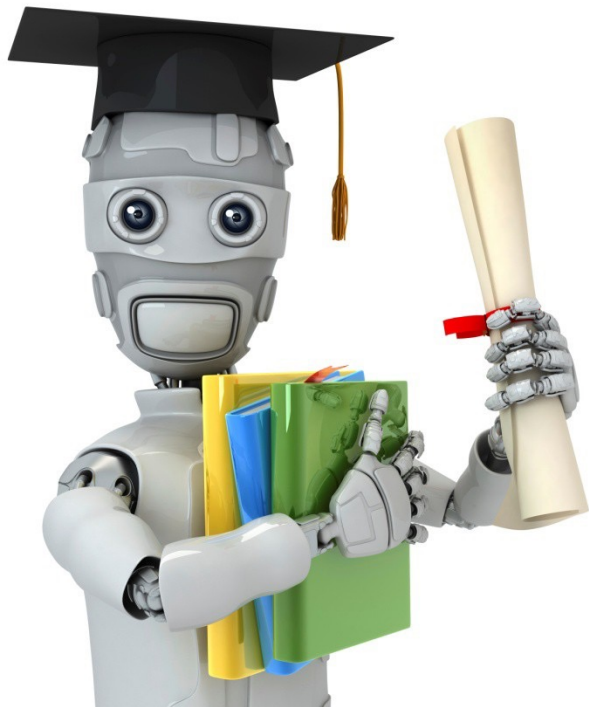


我们怎么表示 h ?

$$h_{\theta}(x) = \theta_0 + \theta_1 x \quad f_{w,b}(x) = wx + b$$



单变量线性回归。



机器学习

单变量线性回归

代价函数

训练集

Size in feet ² (x)	Price (\$) in 1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

训练样本数
m=47

假设: $h_{\theta}(x) = \theta_0 + \theta_1 x$

θ_i : 参数

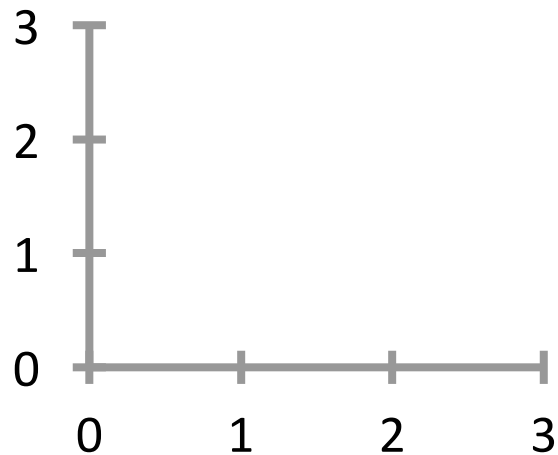
如何选择 θ_i ?

m<2: 欠定方程

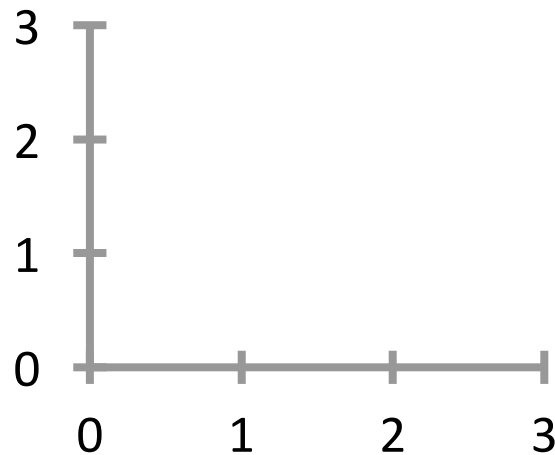
m=2: 适定方程

m>2: 超定方程

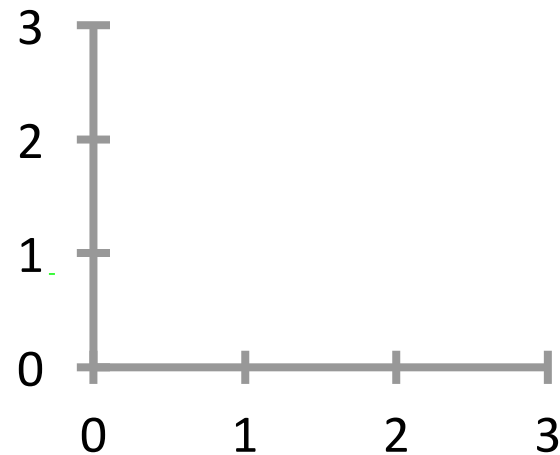
$$h_{\theta}(x) = \theta_0 + \theta_1 x$$



$$\theta_0 = 1.5$$
$$\theta_1 = 0$$

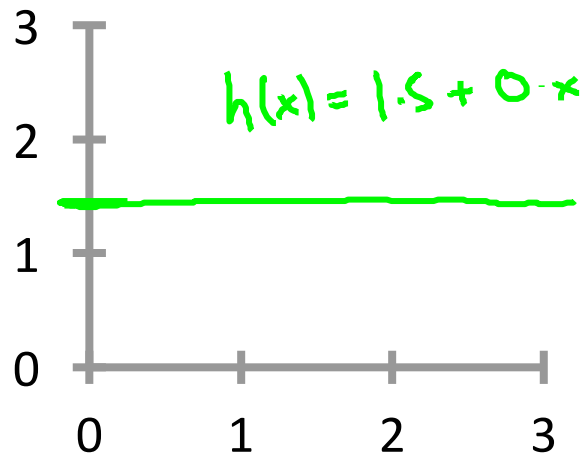


$$\theta_0 = 0$$
$$\theta_1 = 0.5$$



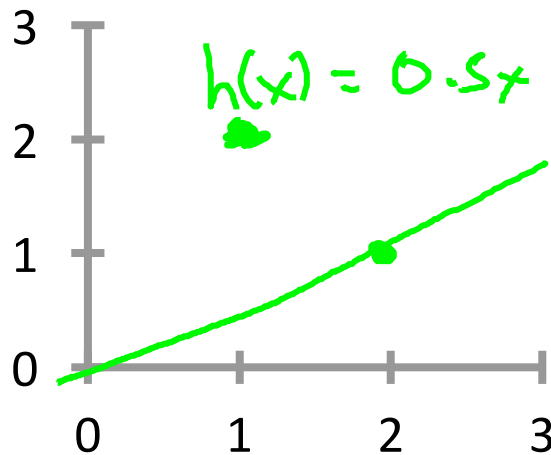
$$\theta_0 = 1$$
$$\theta_1 = 0.5$$

$$\underline{h_{\theta}(x)} = \theta_0 + \theta_1 x$$



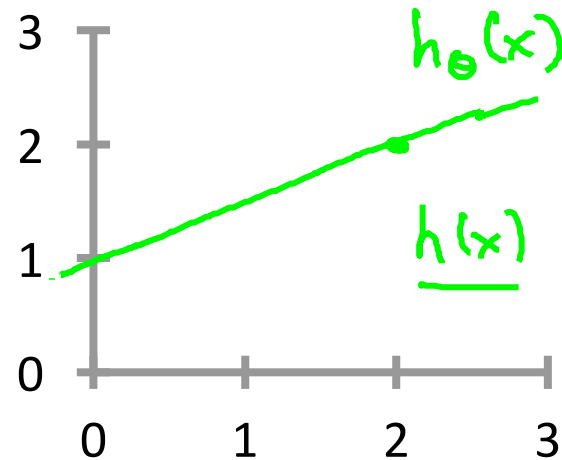
$$\rightarrow \theta_0 = 1.5$$

$$\rightarrow \theta_1 = 0$$



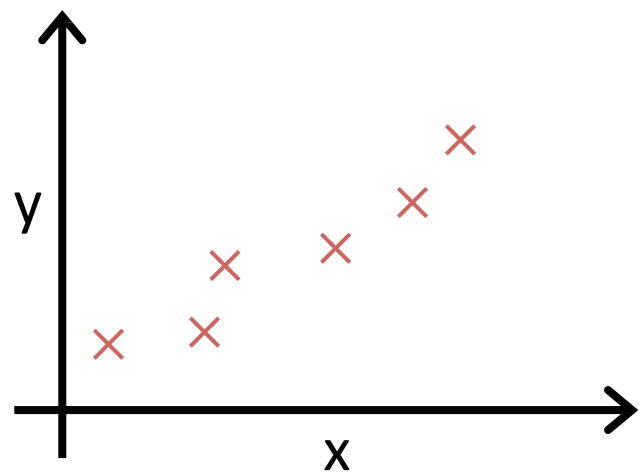
$$\rightarrow \theta_0 = 0$$

$$\rightarrow \theta_1 = 0.5$$



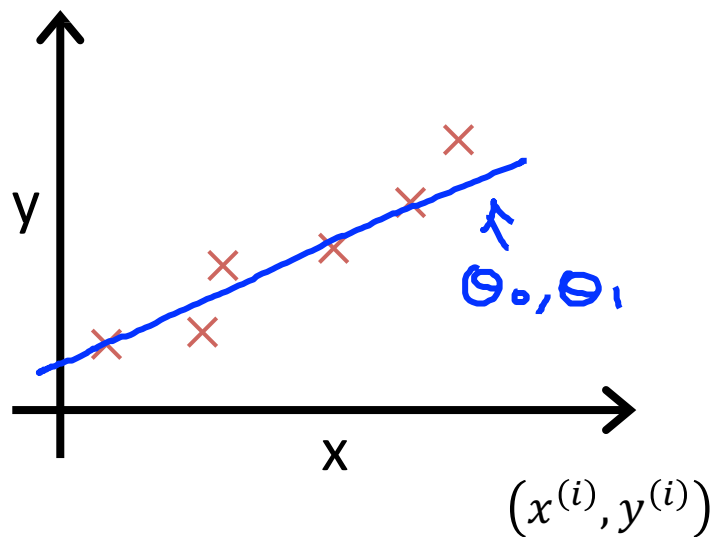
$$\rightarrow \theta_0 = 1$$

$$\rightarrow \theta_1 = 0.5$$



训练样本
 (x, y)

选择 θ_0, θ_1 使 $h_{\theta}(x)$ 逼近 y



选择 θ_0, θ_1 使 $h_{\theta}(x)$ 逼近 y

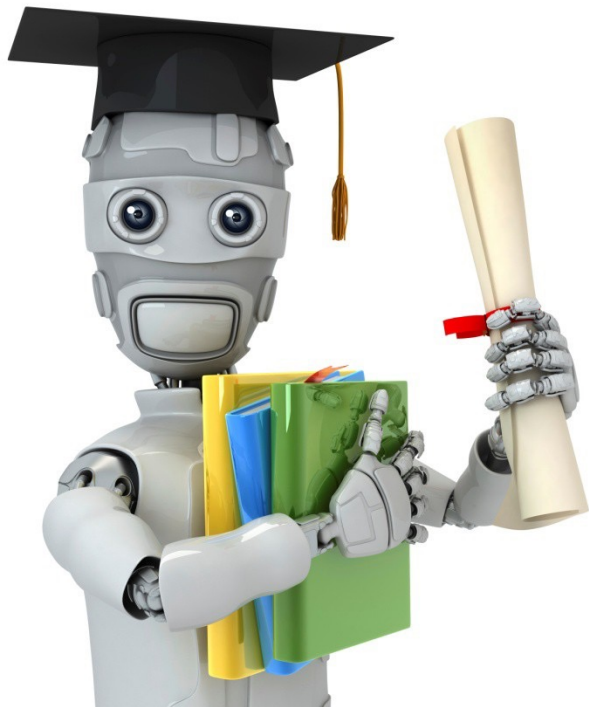
$$\underset{\theta_0, \theta_1}{\text{minimize}} \frac{1}{2m} \sum_{i=1}^m \underbrace{(h_{\theta}(x^{(i)}) - y^{(i)})^2}_{h_{\theta}(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}}$$

训练样本数 m

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$\underset{\theta_0, \theta_1}{\text{minimize}} \underbrace{J(\theta_0, \theta_1)}$$

代价函数 Cost Function



机器学习

单变量线性回归

代价函数 直观理解I

假设:

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

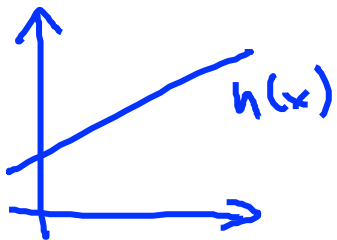
参数:

$$\theta_0, \theta_1$$

代价函数:

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

目标: minimize $J(\theta_0, \theta_1)$
 θ_0, θ_1

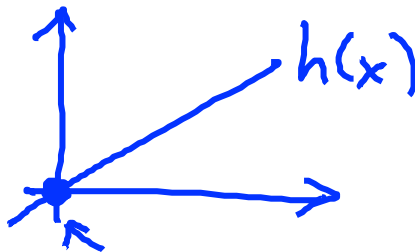


简化

$$h_{\theta}(x) = \theta_1 x$$

$$\theta_0 = 0$$

$$\theta_1$$



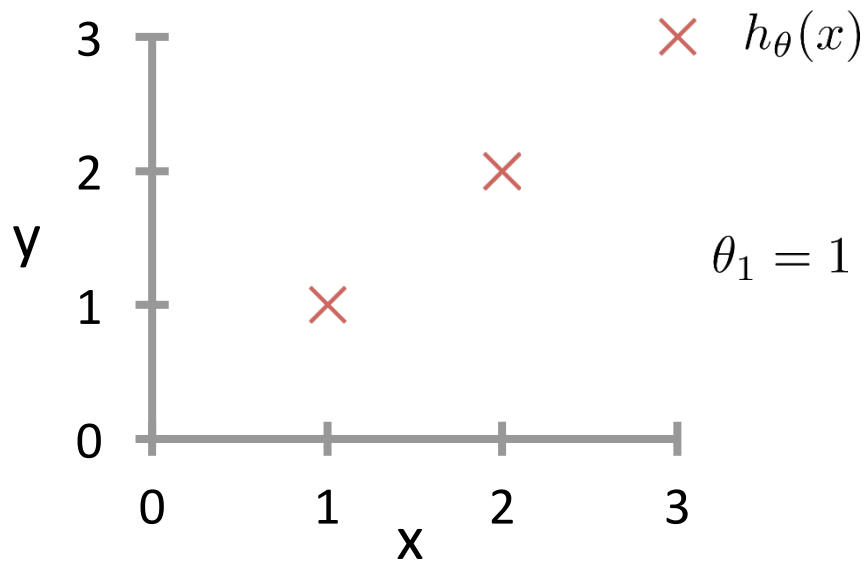
$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

minimize $J(\theta_1)$
 θ_1

$$\theta_1, x^{(i)}$$

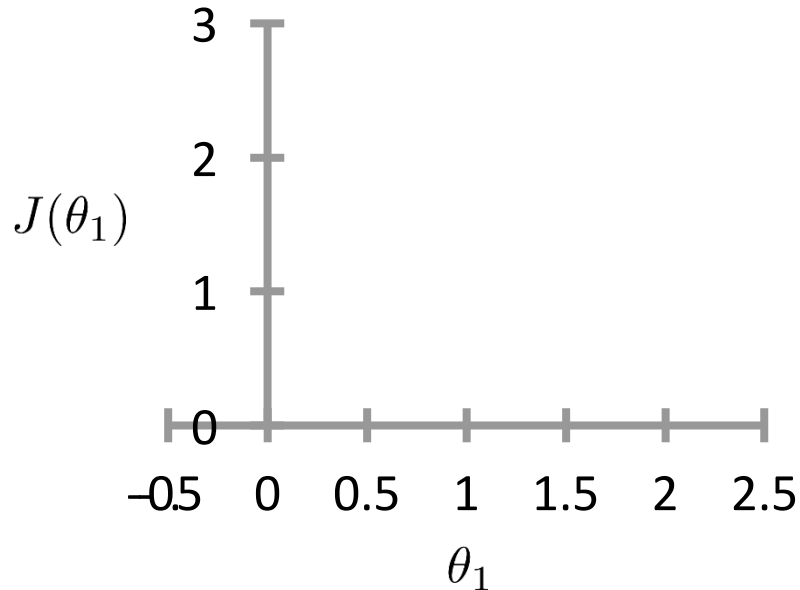
$$h_{\theta}(x)$$

(对于固定 θ_1 , 这是 x 的函数)



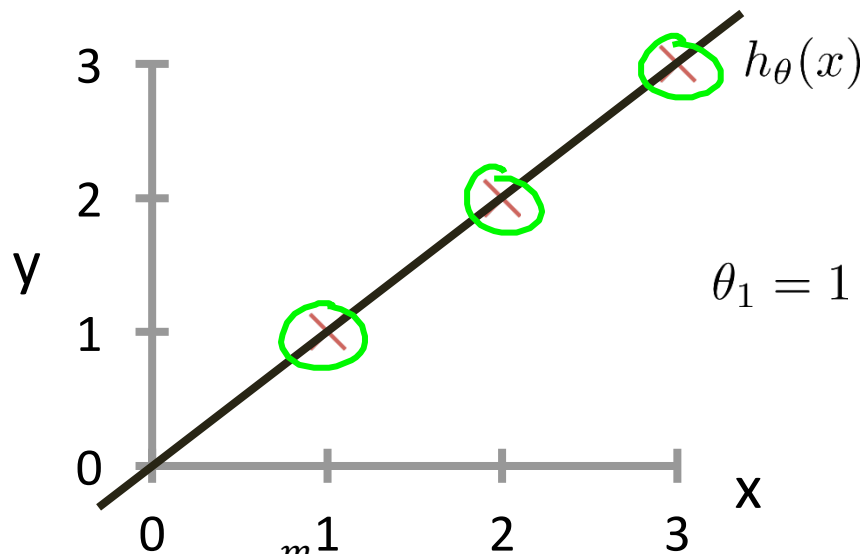
$$J(\theta_1)$$

(代价函数为 θ_1 的函数)



$$h_{\theta}(x)$$

(对于固定 θ_1 , 这是 x 的函数)

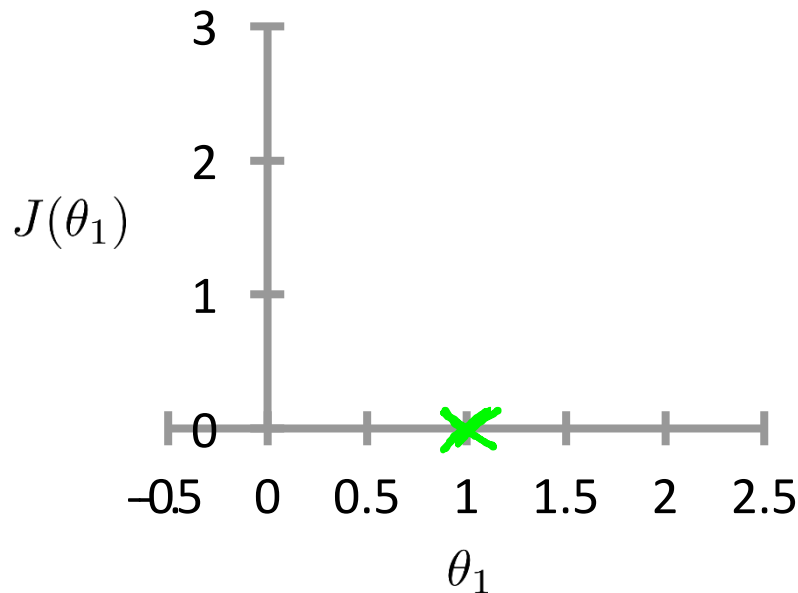


$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$= \frac{1}{2m} \sum_{i=1}^m (\theta_1 x^{(i)} - y^{(i)})^2$$

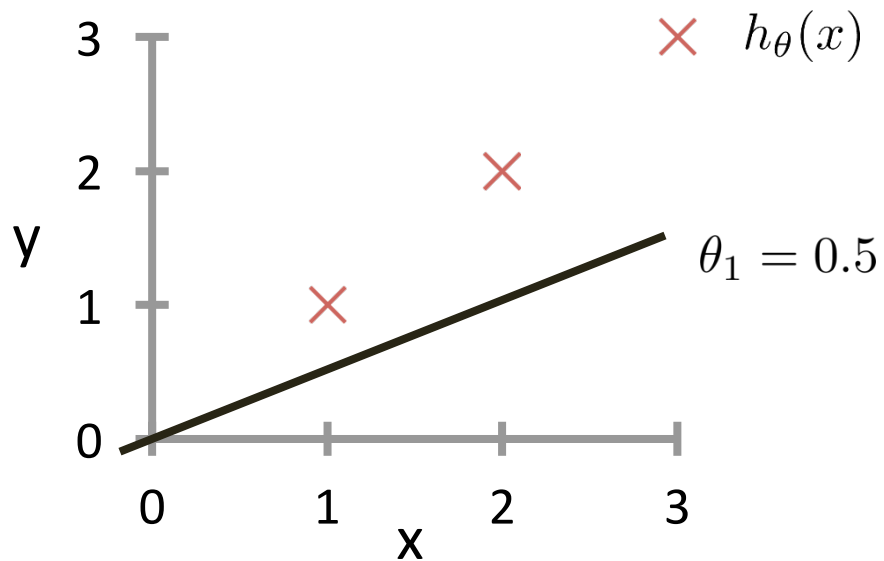
$$J(\theta_1)$$

(代价函数为 θ_1 的函数)



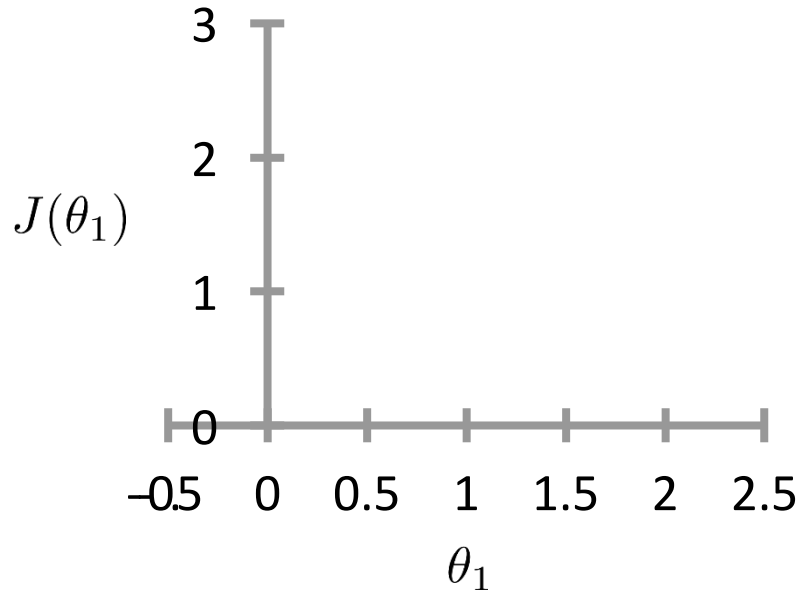
$$h_{\theta}(x)$$

(for fixed θ_1 , this is a function of x)



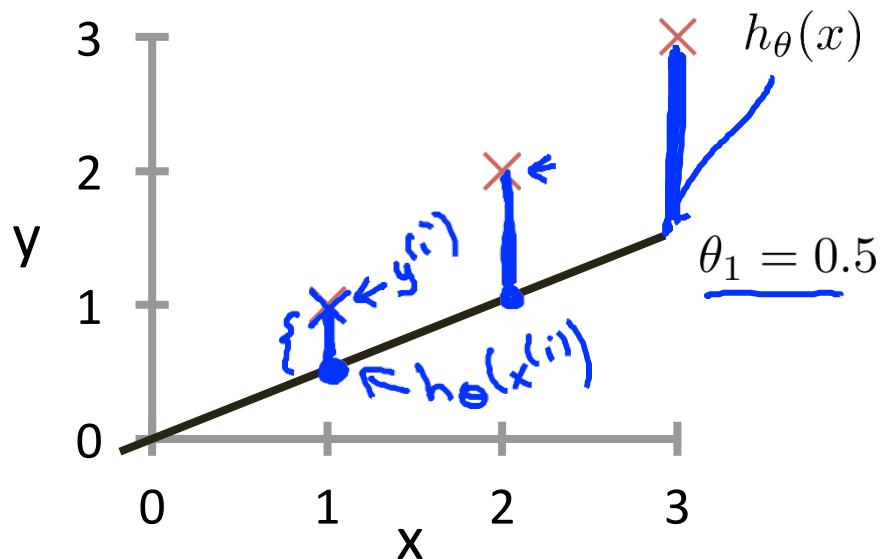
$$J(\theta_1)$$

(function of the parameter θ_1)



$$h_{\theta}(x)$$

(for fixed θ_1 , this is a function of x)

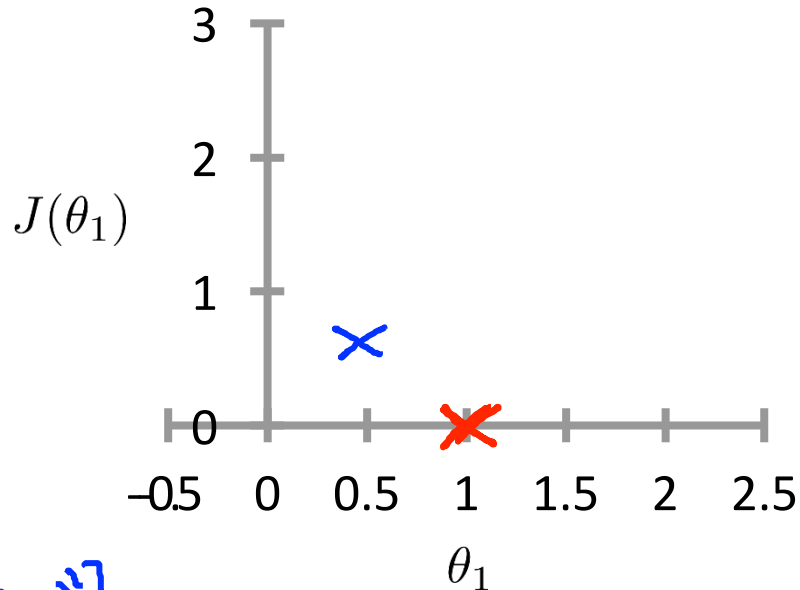


$$J(0.5) = \frac{1}{2m} [(0.5-1)^2 + (1-2)^2 + (1.5-3)^2]$$

$$= \frac{1}{2 \times 3} (3.5) = \frac{3.5}{6} \approx \underline{0.58}$$

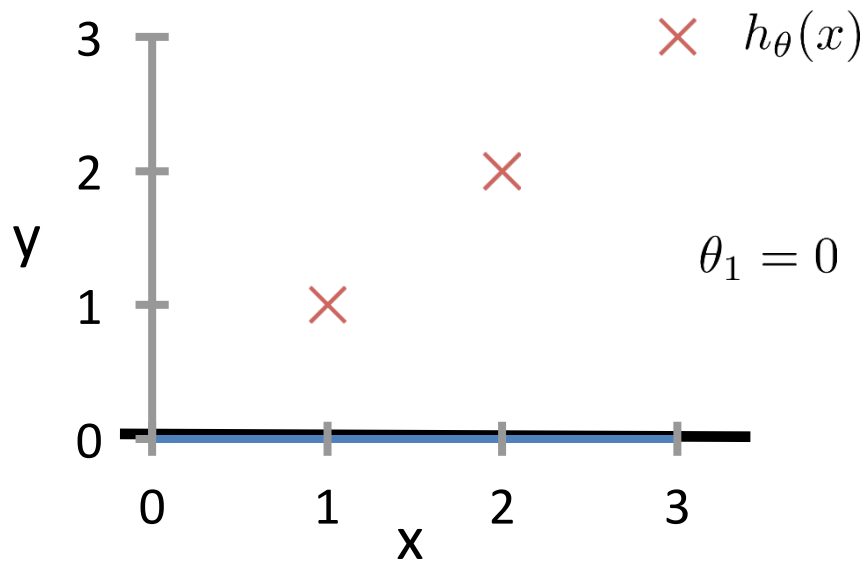
$$J(\theta_1)$$

(function of the parameter θ_1)



$$h_{\theta}(x)$$

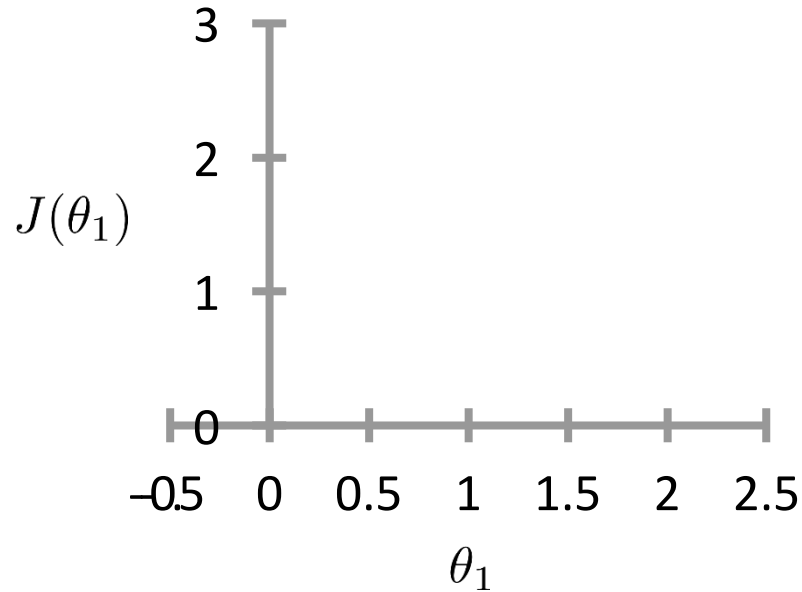
(for fixed θ_1 , this is a function of x)



$$J(\theta_1)$$

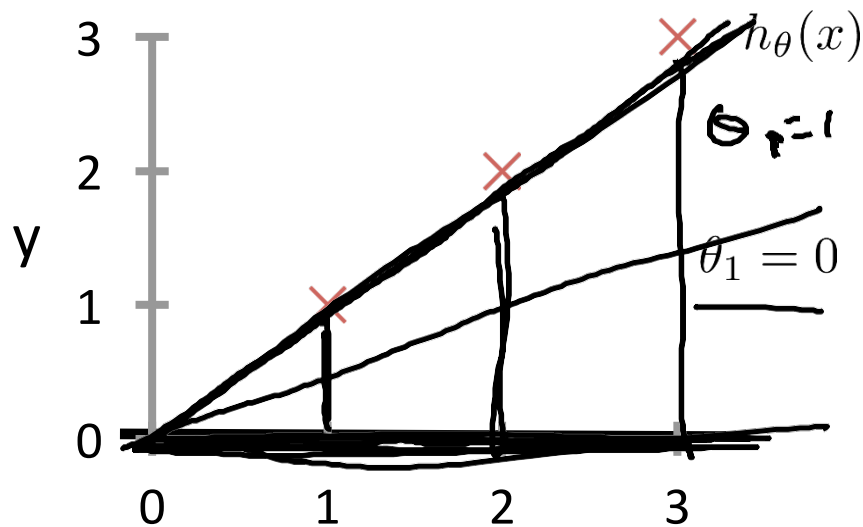
↗

(function of the parameter θ_1)

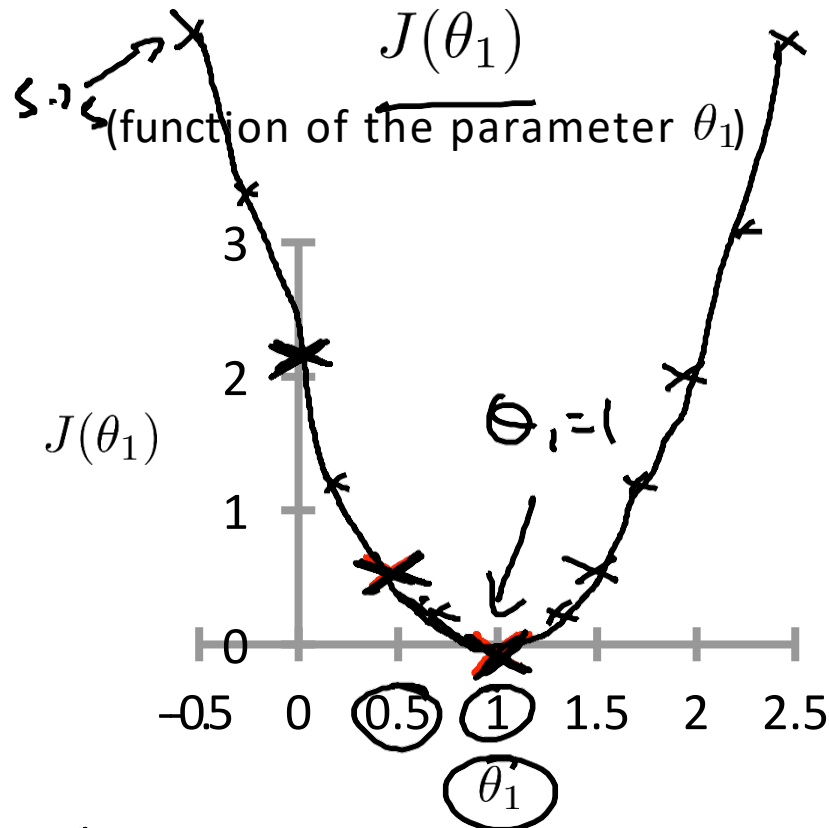


$$h_{\theta}(x)$$

(for fixed θ_1 , this is a function of x)

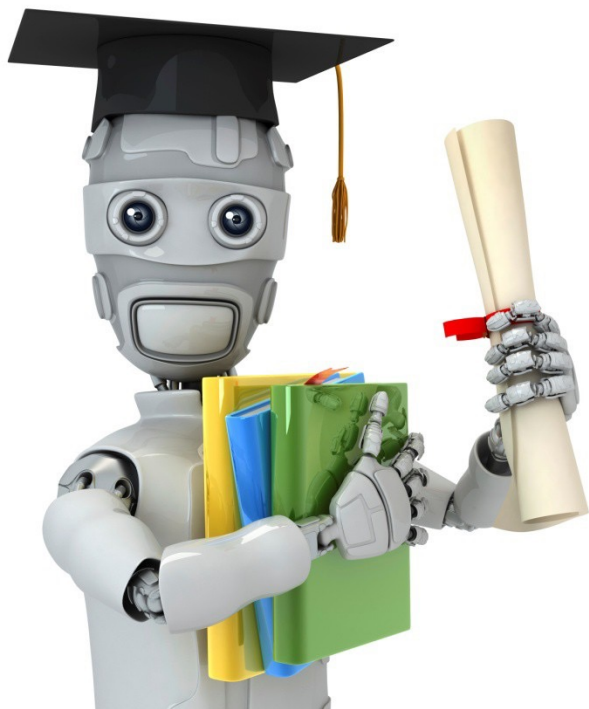


$$J(0) = \frac{1}{2m} (1^2 + 2^2 + 3^2) = \frac{1}{6} \cdot 14 \approx 2.3$$



$$h(x) = -0.5x$$

minimize $J(\theta_1)$



机器学习

单变量线性回归

代价函数 直观理解II

假设: $h_{\theta}(x) = \theta_0 + \theta_1 x$

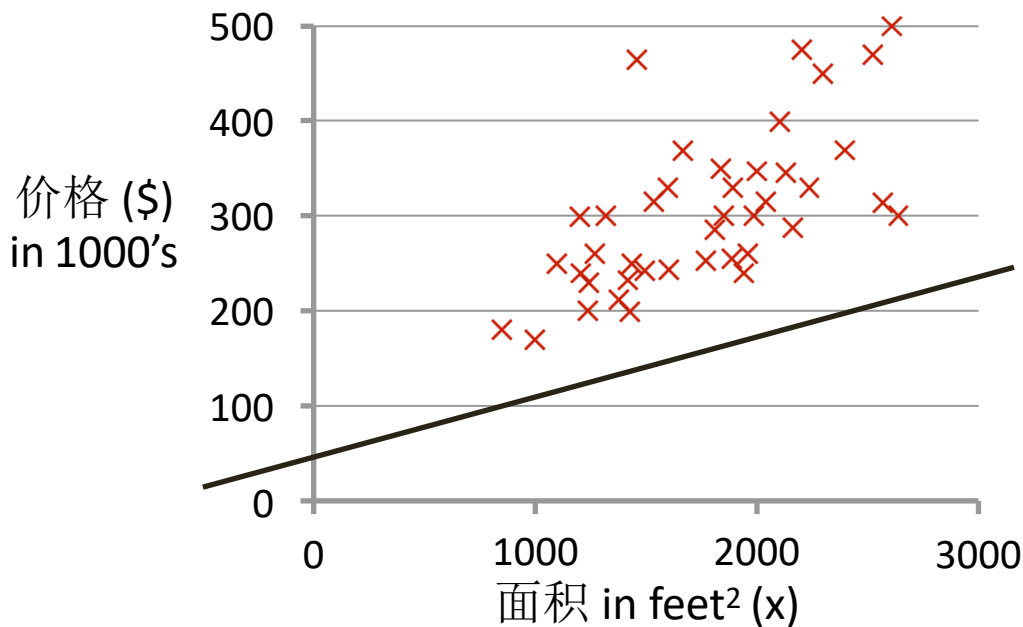
参数: θ_0, θ_1

代价函数: $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$

目标: $\underset{\theta_0, \theta_1}{\text{minimize}} J(\theta_0, \theta_1)$

$$h_{\theta}(x)$$

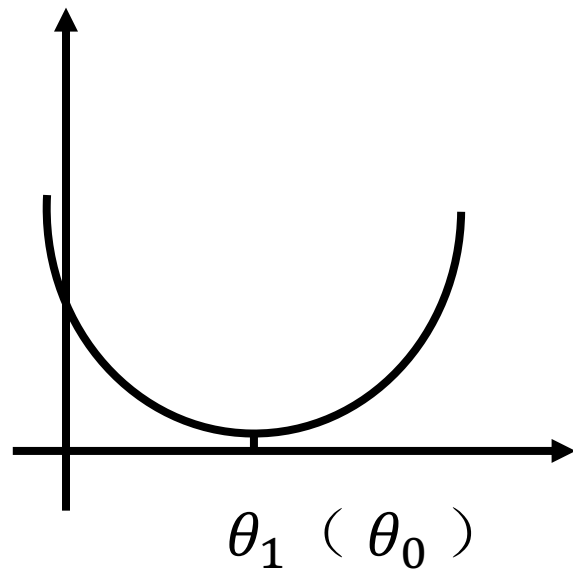
(对于固定的 θ_0, θ_1 , 为 x 的函数)



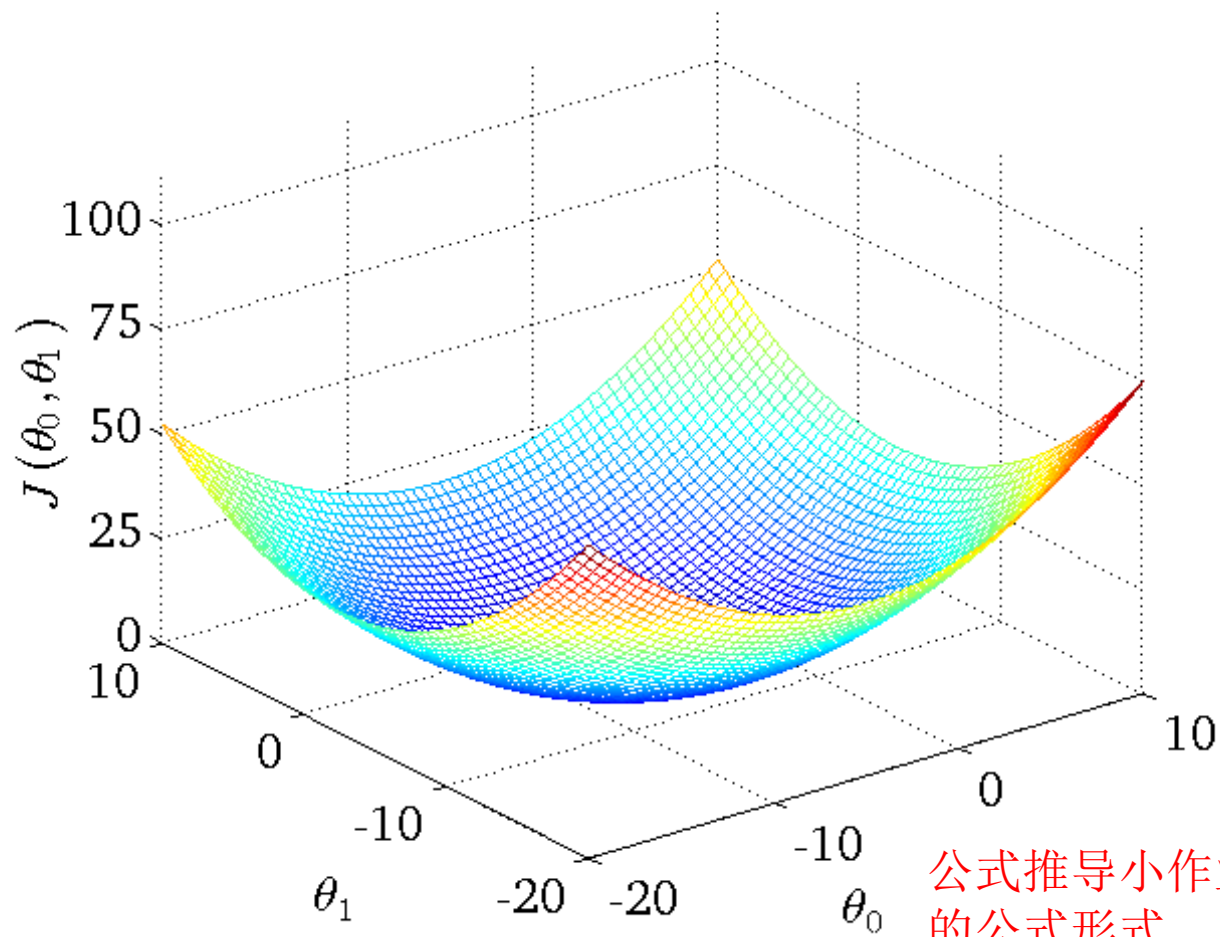
$$h_{\theta}(x) = 50 + 0.06x$$

$$J(\theta_0, \theta_1)$$

(函数中的参数 θ_0, θ_1)



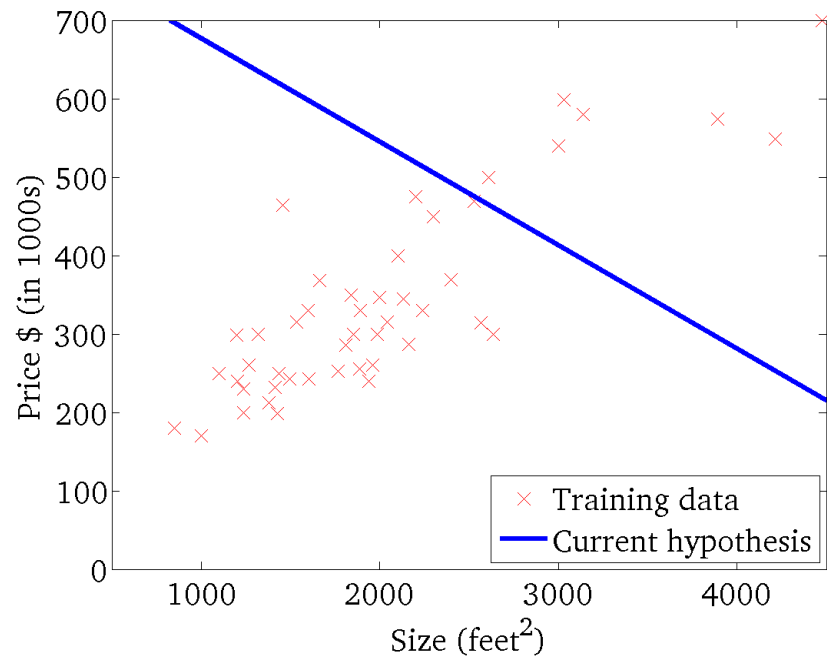
公式推导小作业1-1: 根据上页的假设,
证明 $J(\theta_0)$ 和 $J(\theta_1)$ 的形状为抛物线和
凹弧



公式推导小作业1-2：写出 $J(\theta_0, \theta_1)$ 的公式形式

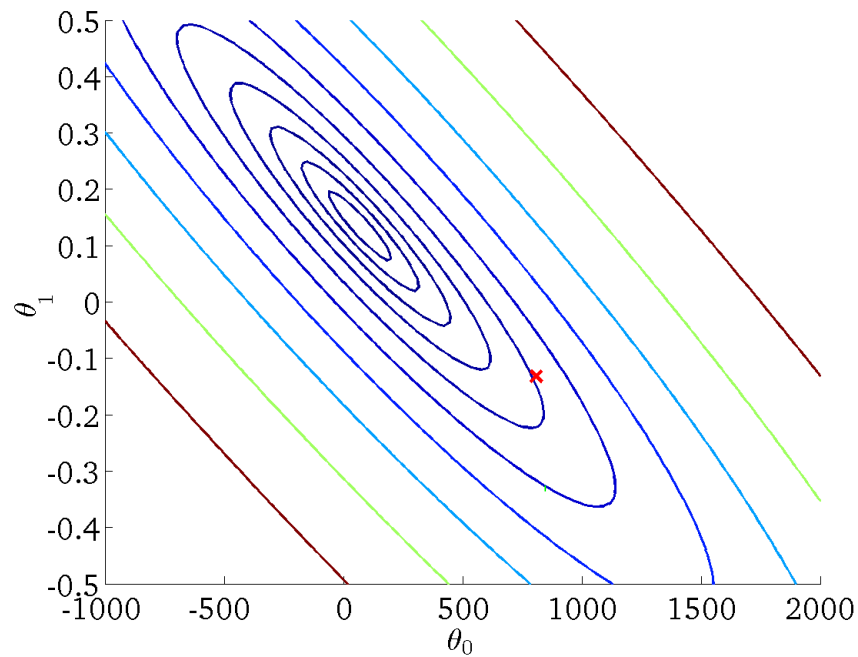
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)

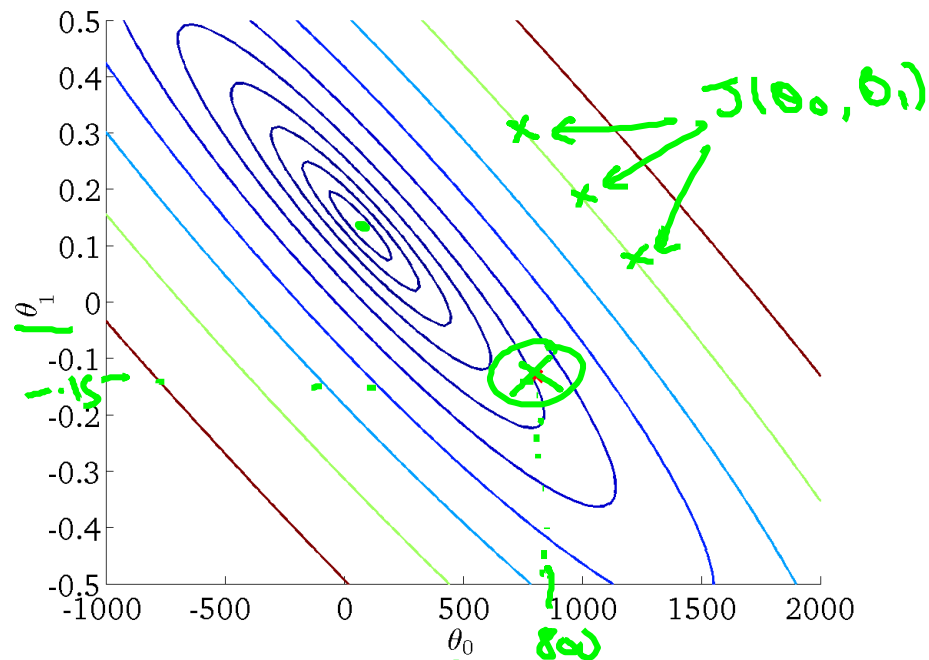
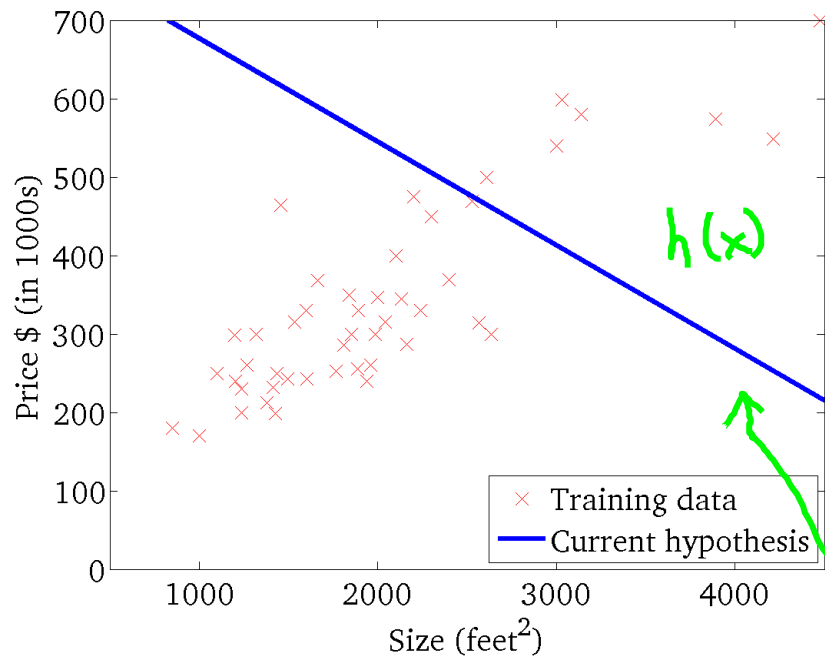


$$h_{\theta}(x)$$

$$J(\theta_0, \theta_1)$$

(for fixed θ_0, θ_1 , this is a function of x)

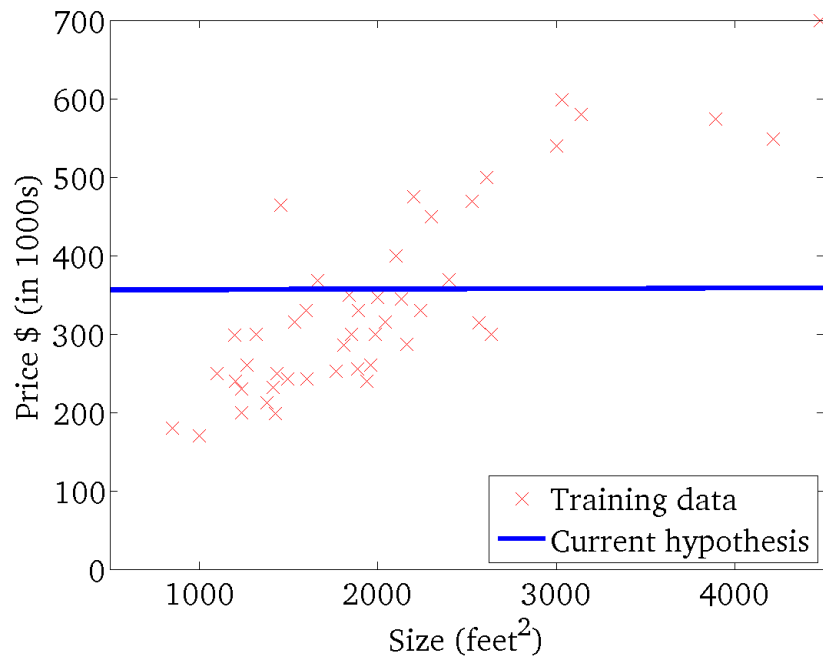
(function of the parameters θ_0, θ_1)



θ_0, θ_1

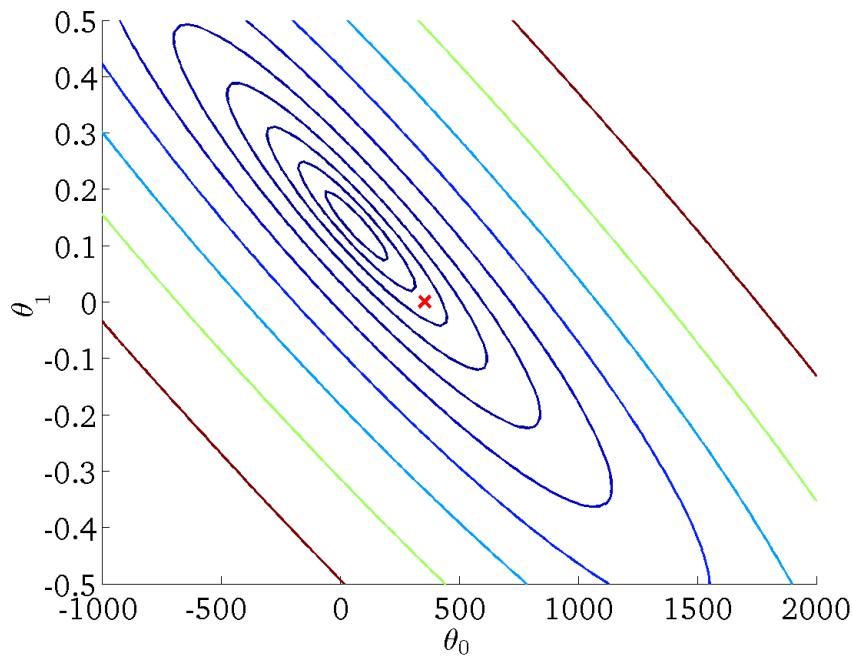
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



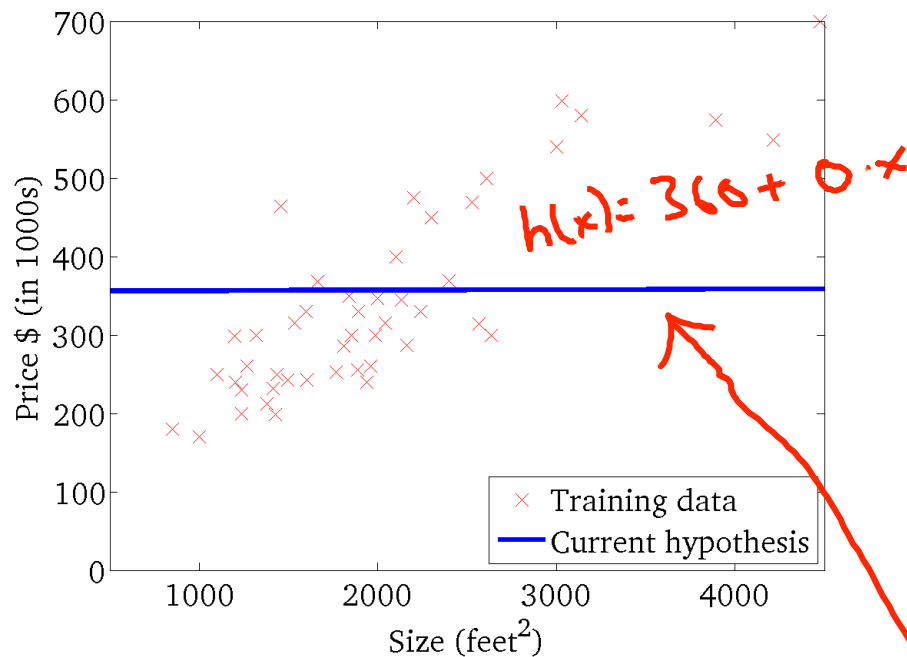
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



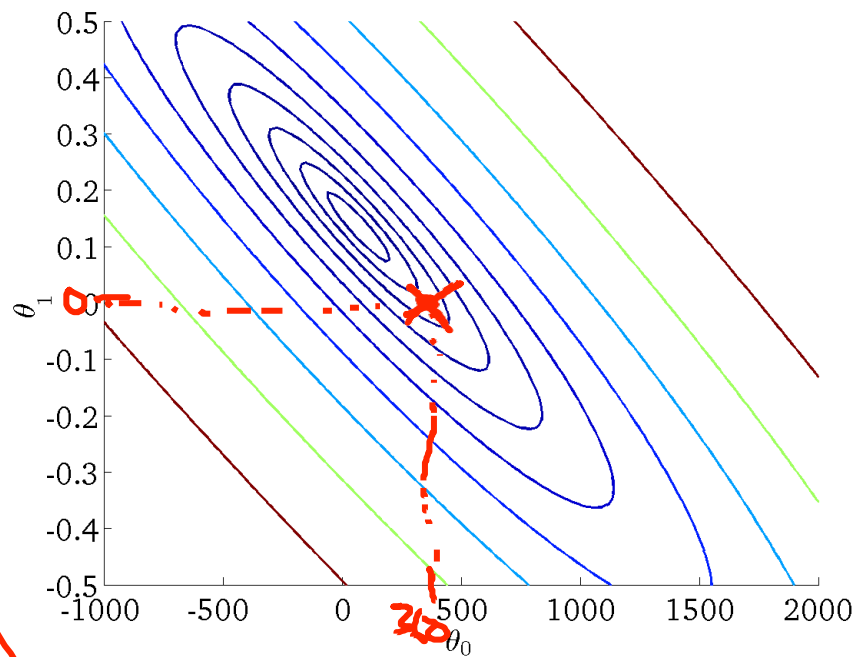
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

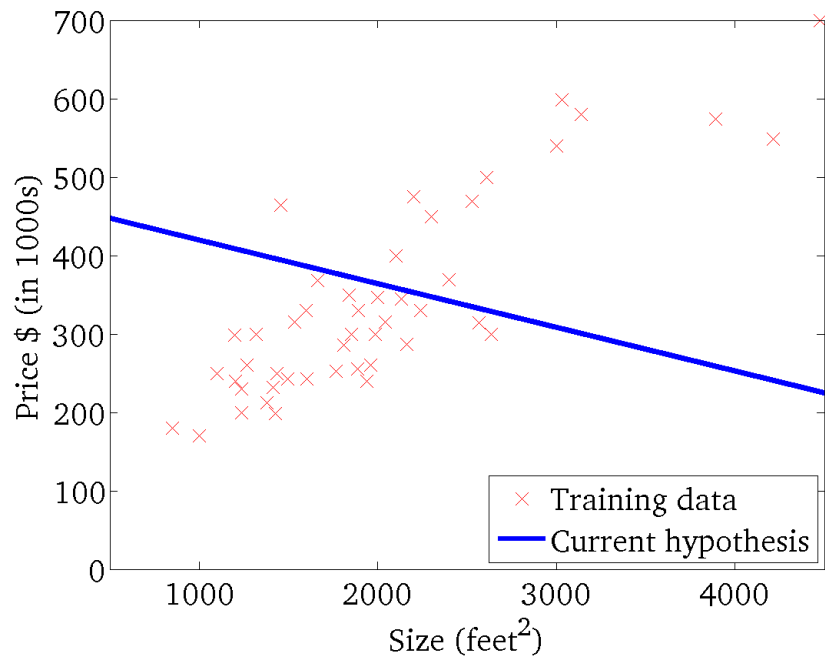
(function of the parameters θ_0, θ_1)



$$\begin{cases} \theta_0 = 360 \\ \theta_1 = 0 \end{cases}$$

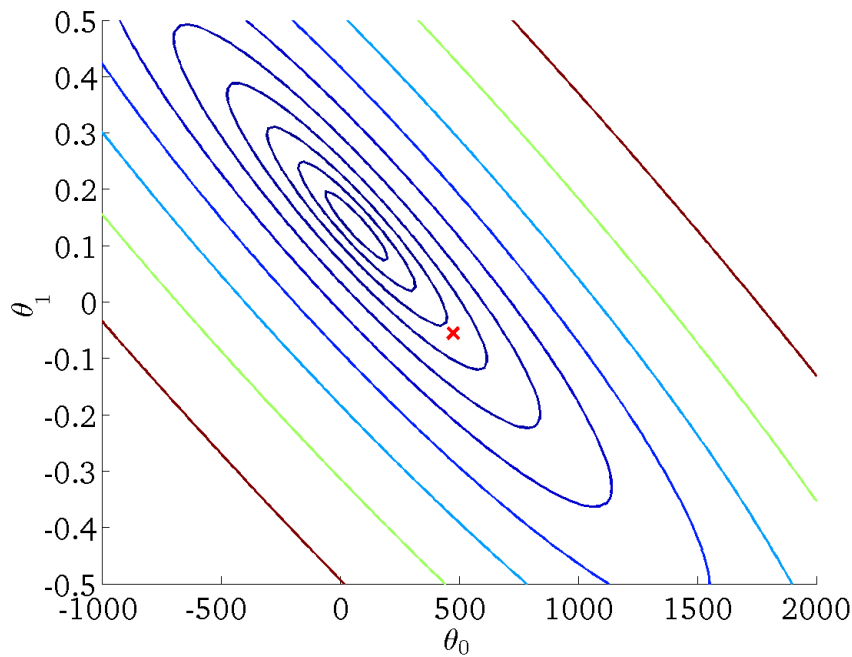
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



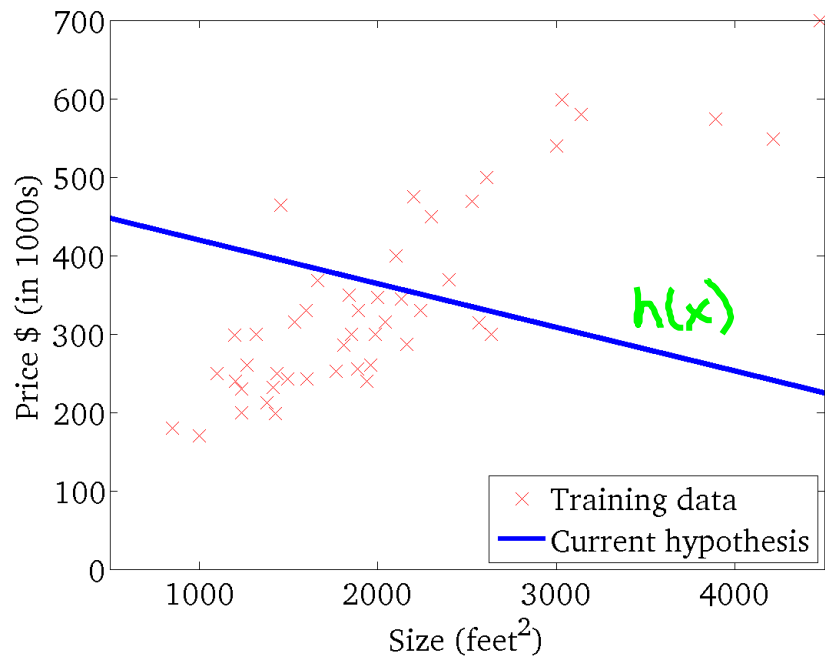
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



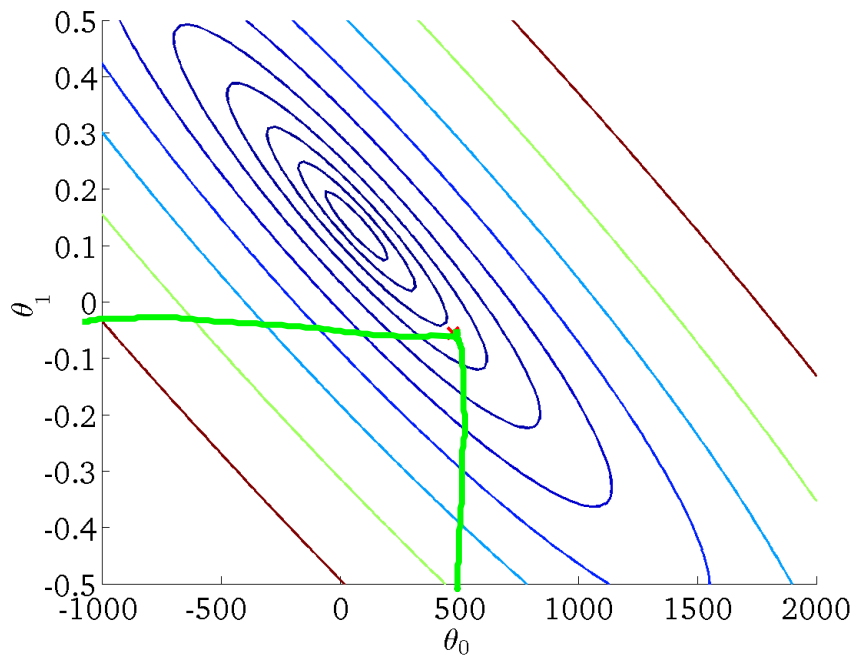
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



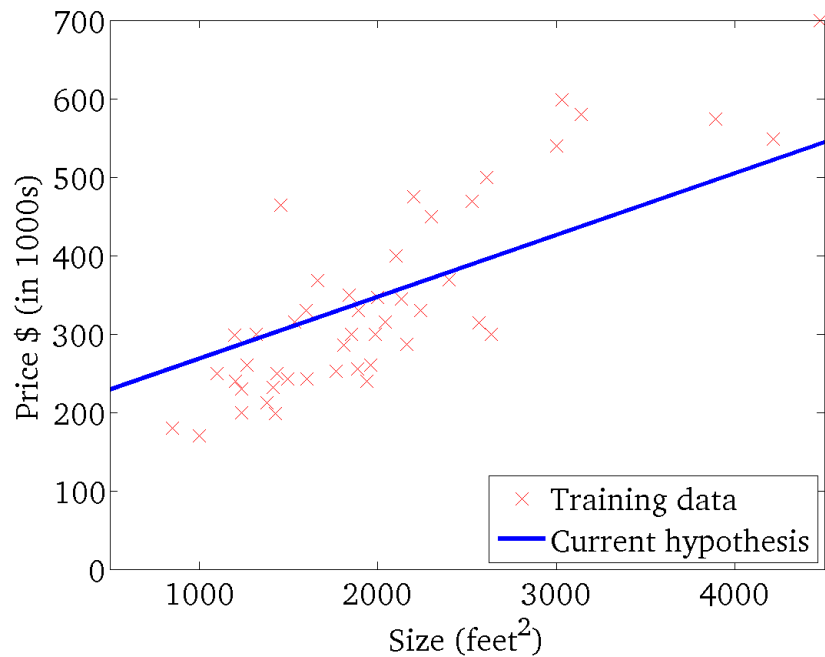
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



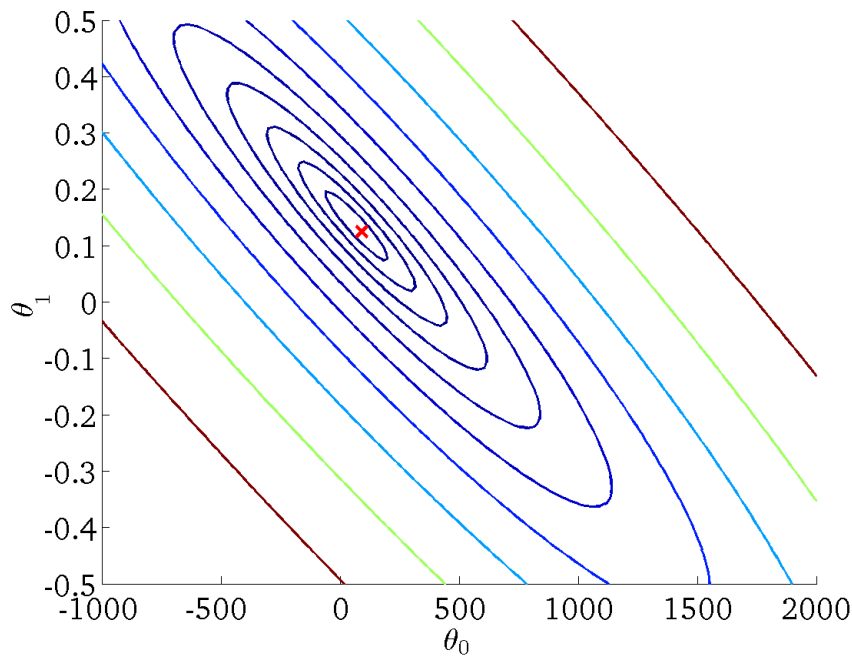
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



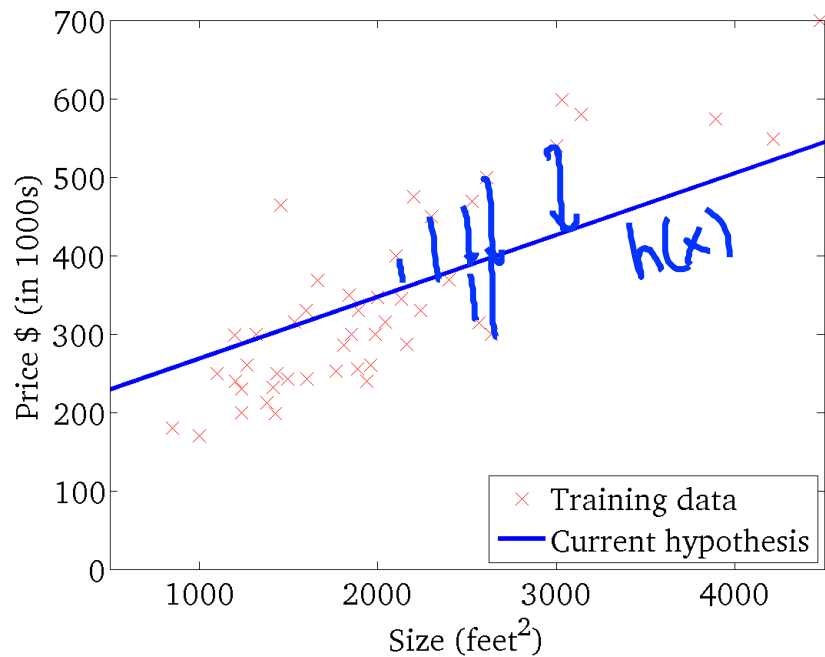
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



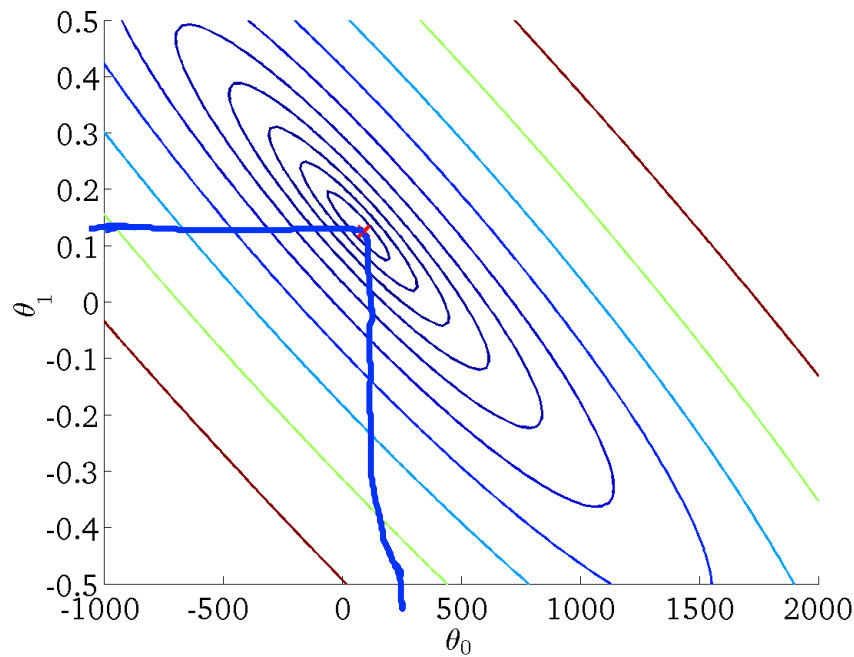
$$h_{\theta}(x)$$

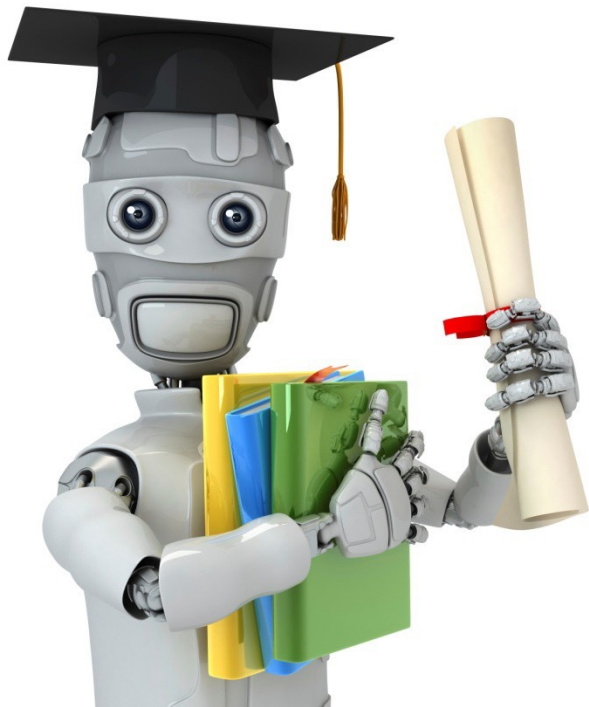
(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)





机器学习

单变量线性回归

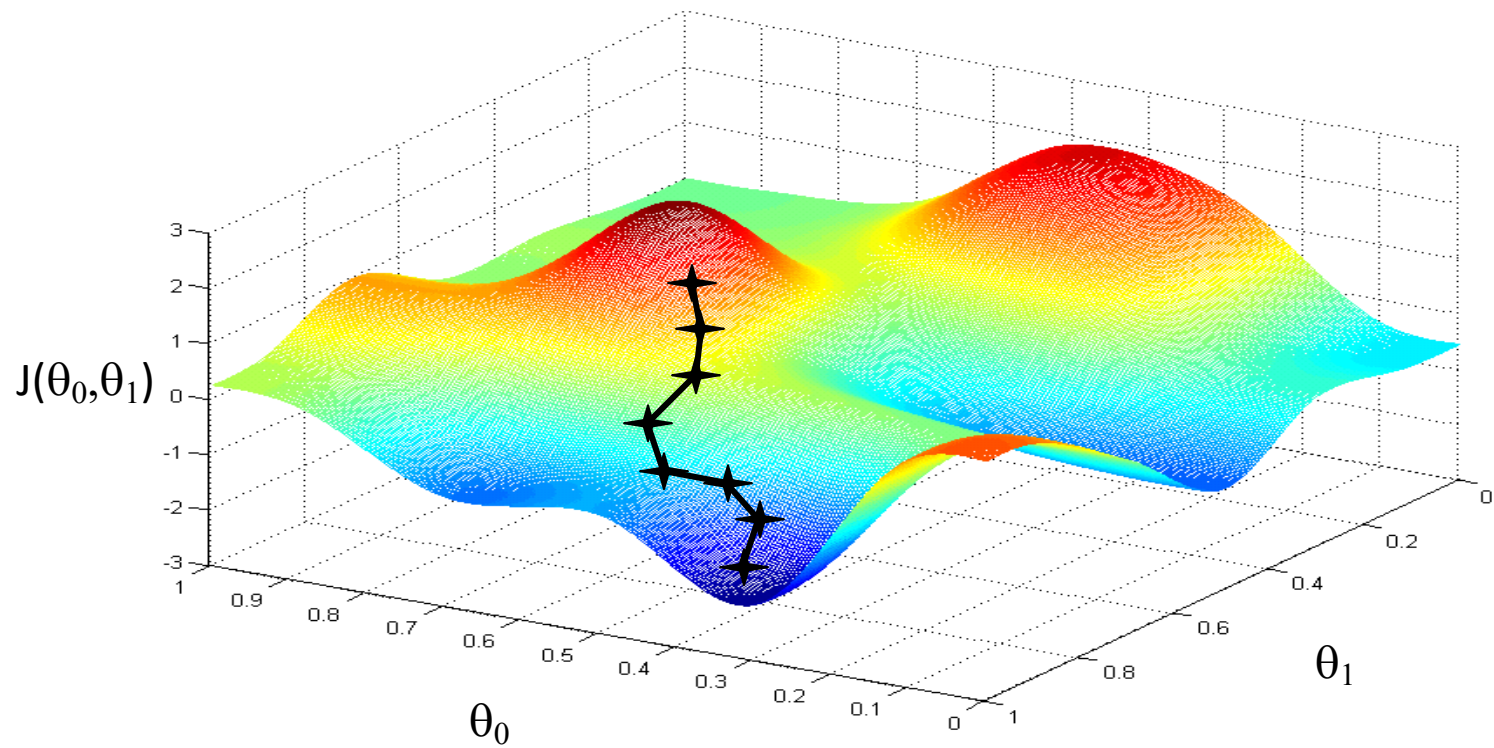
梯度下降

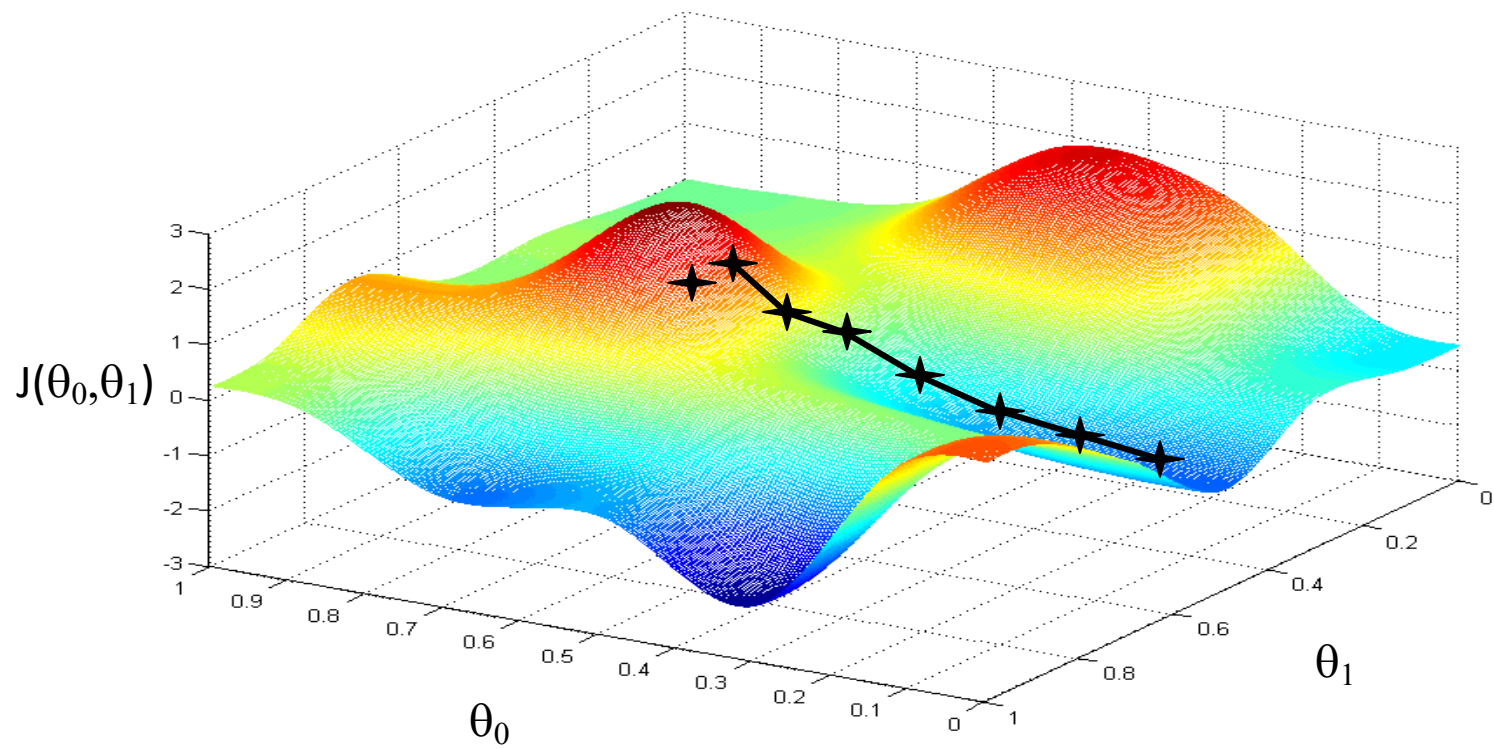
根据函数 $J(\theta_0, \theta_1)$

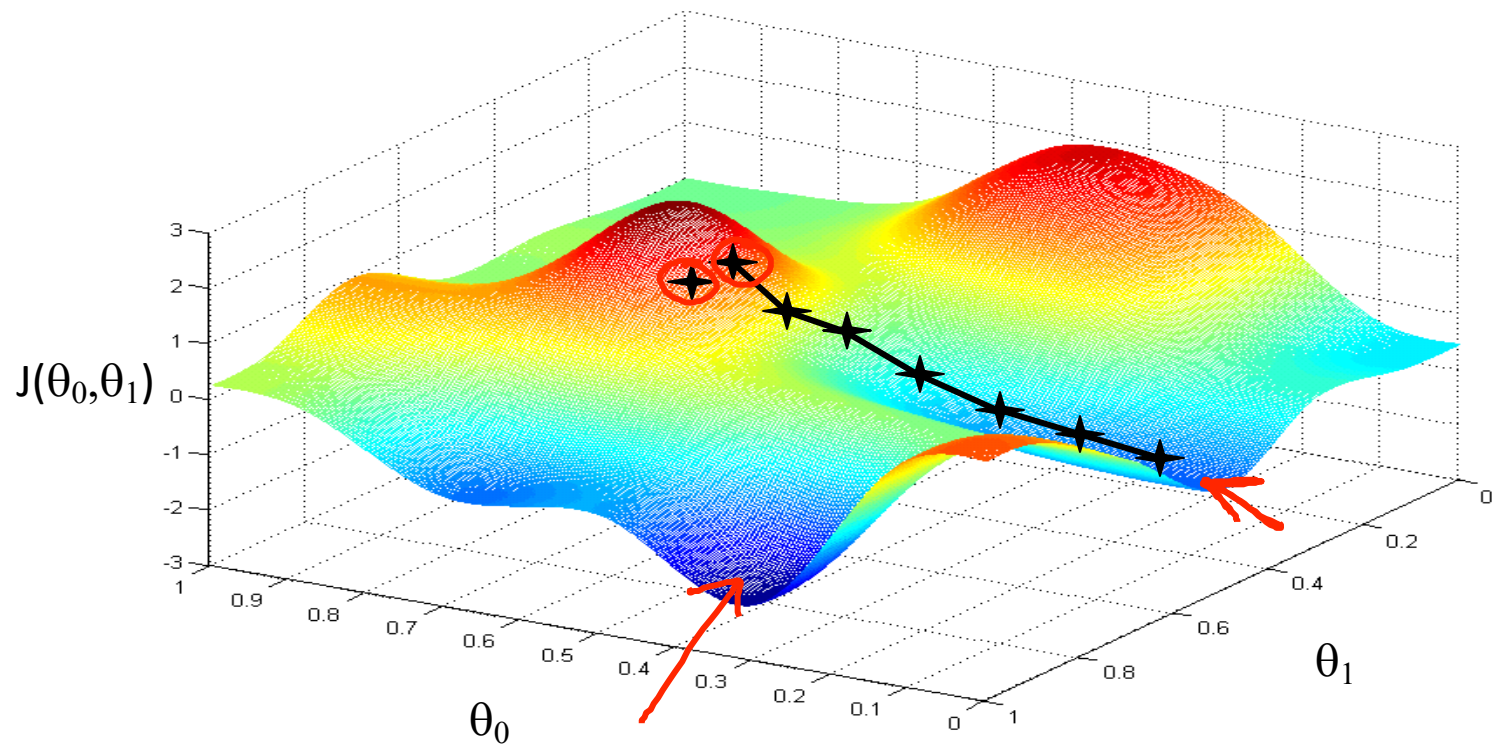
想要求得 $\min_{\theta_0, \theta_1} J(\theta_0, \theta_1)$

算法框架:

- 预设 θ_0, θ_1
- 不停改变 θ_0, θ_1 使 $J(\theta_0, \theta_1)$ 减小
直到得到我们希望的最小值







梯度下降算法

repeat until convergence {
 $\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$ (for $j = 0$ and $j = 1$)
}

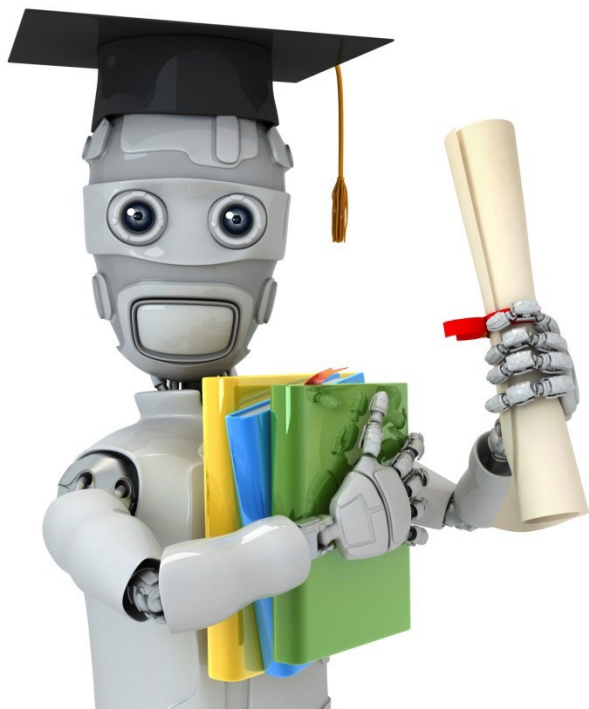
 学习率

正确：同步更新

```
temp0 :=  $\theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$   
temp1 :=  $\theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$   
 $\theta_0 :=$  temp0  
 $\theta_1 :=$  temp1
```

错误：

```
temp0 :=  $\theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$   
 $\theta_0 :=$  temp0  
temp1 :=  $\theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$   
 $\theta_1 :=$  temp1
```



机器学习

单变量线性回归

梯度下降 直观理解

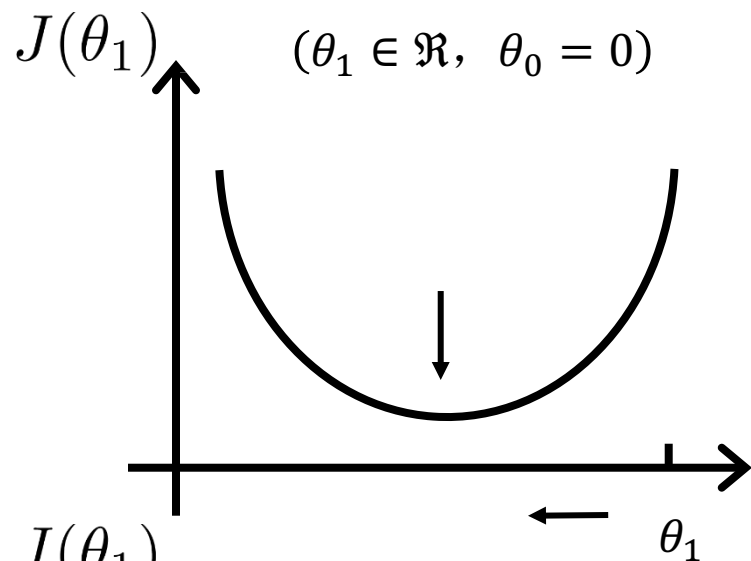
梯度下降算法

repeat until convergence {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

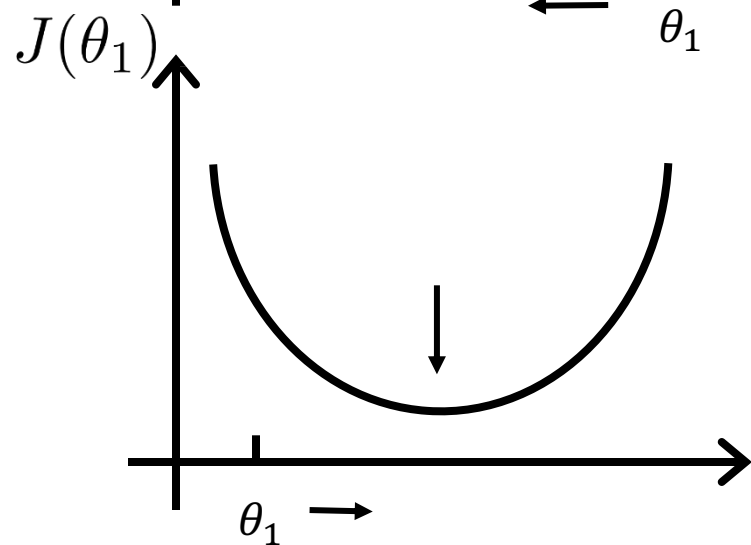
}

(simultaneously update
 $j = 0$ and $j = 1$)



$$\theta_1 := \theta_1 - \alpha \frac{d}{d\theta_1} J(\theta_1)$$

$$\theta_1 := \theta_1 - \alpha(\text{positive number})$$



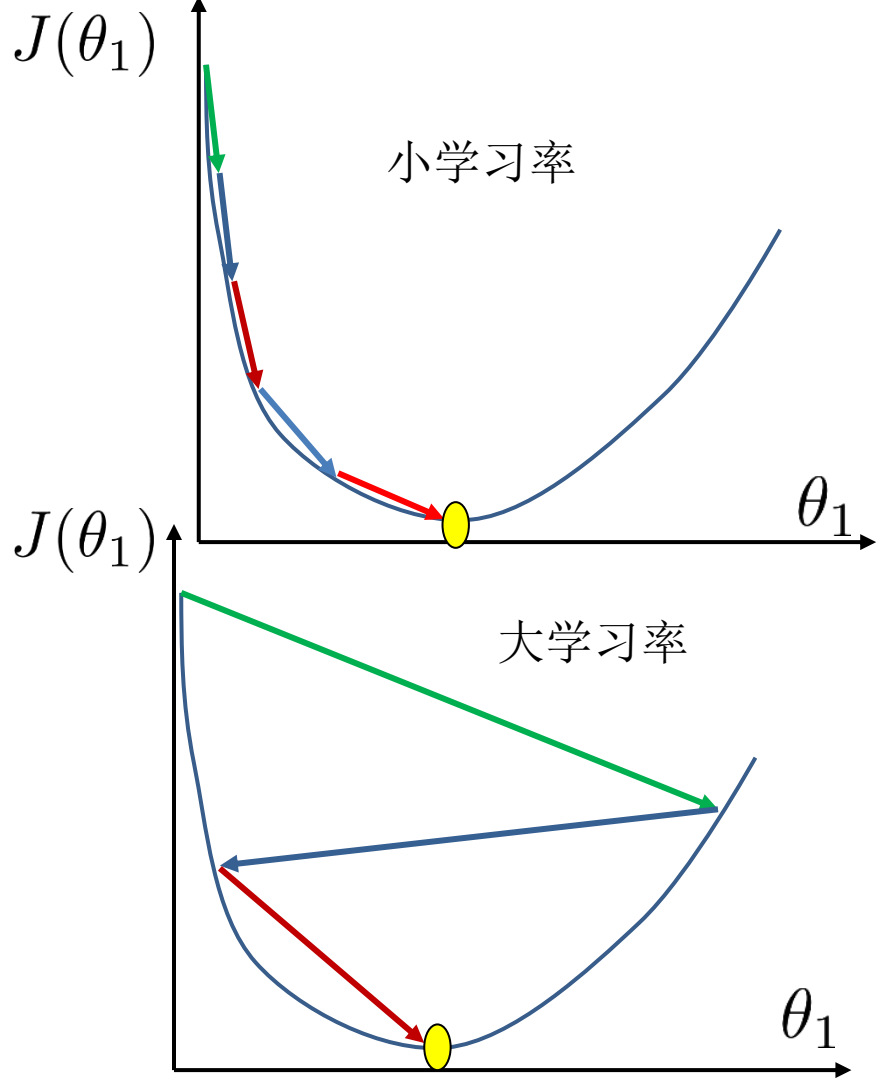
$$\frac{d}{d\theta_1} J(\theta_1) \leq 0$$

$$\theta_1 := \theta_1 - \alpha(\text{negative number})$$

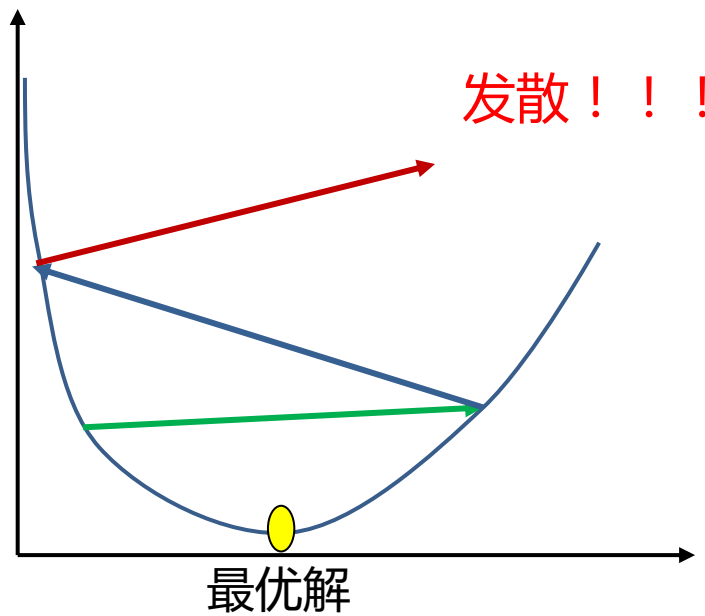
$$\theta_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_1)$$

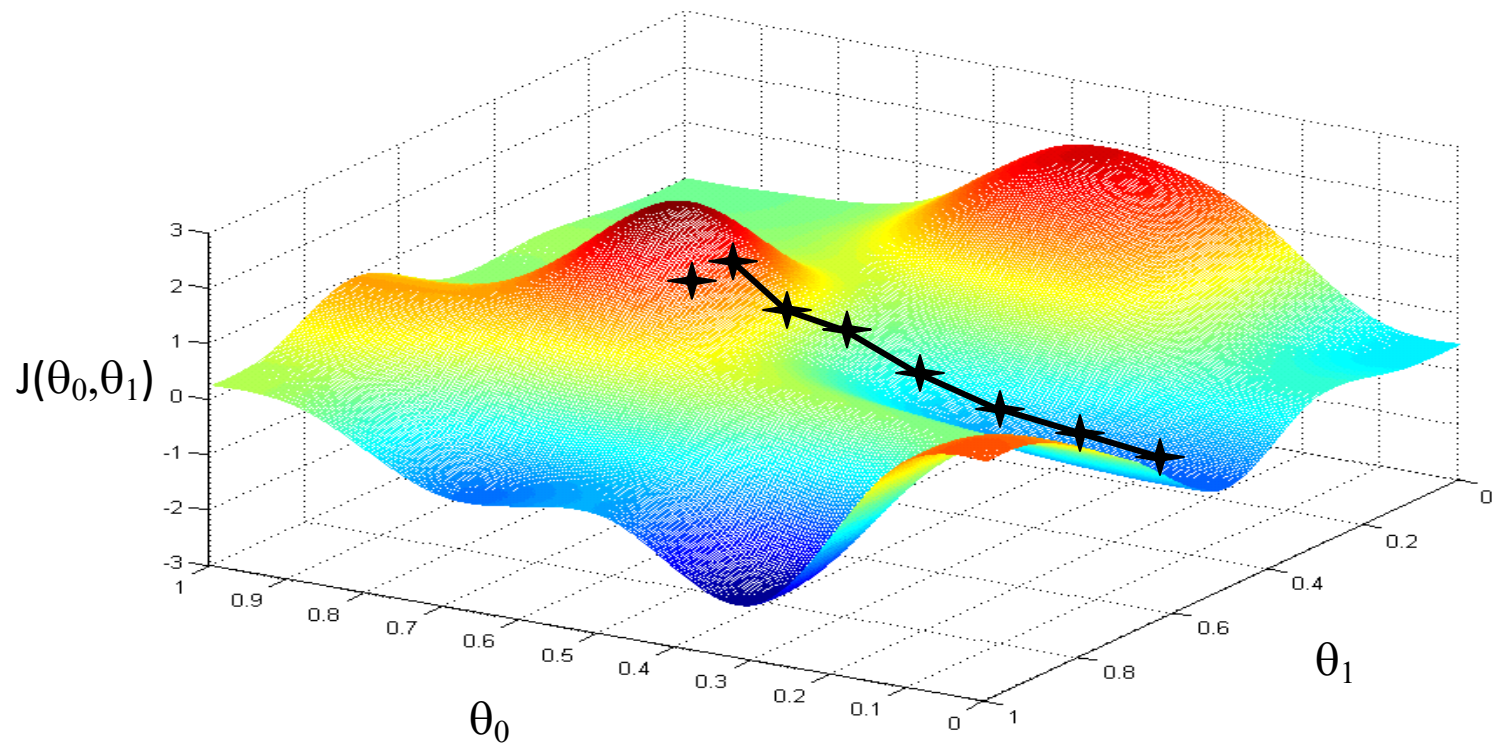
如果 α 极小, 梯度下降就会变慢.

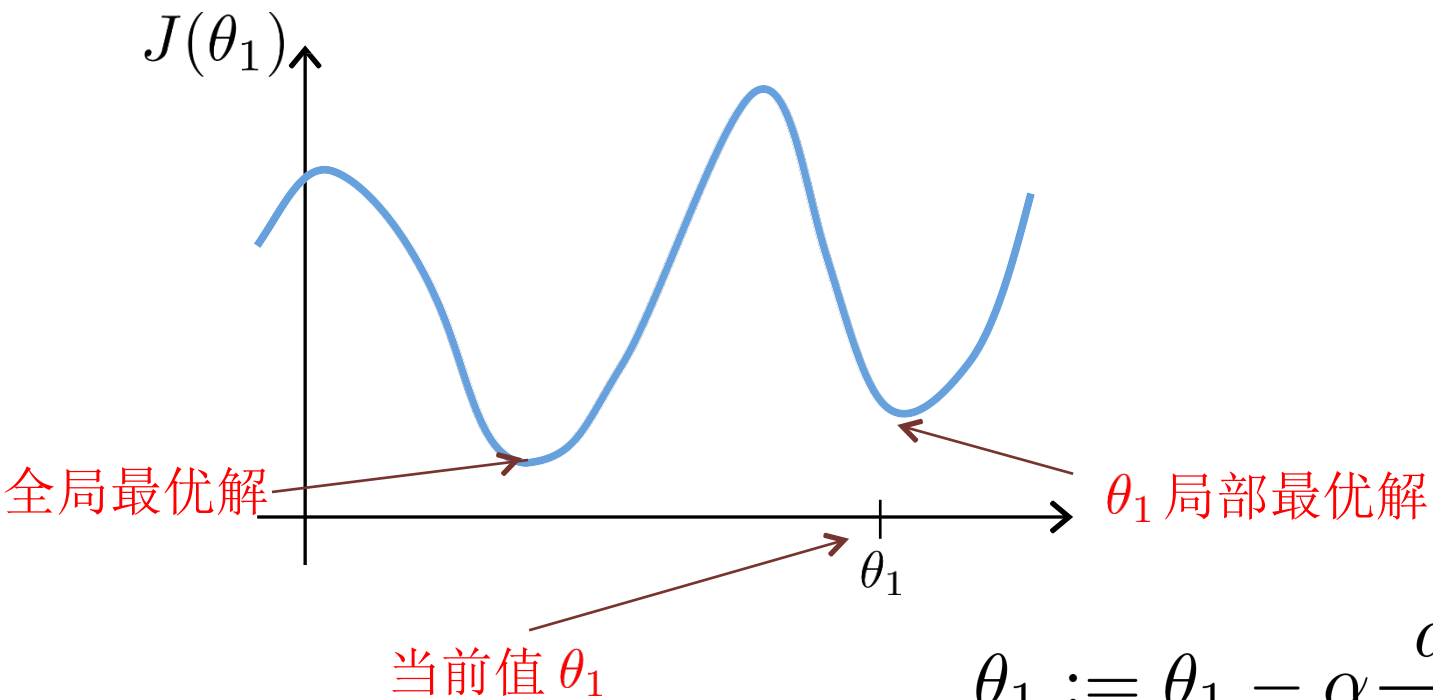
如果 α 极大, 梯度下降很快.



但是，过大的学习率可能
不收敛甚至发散。





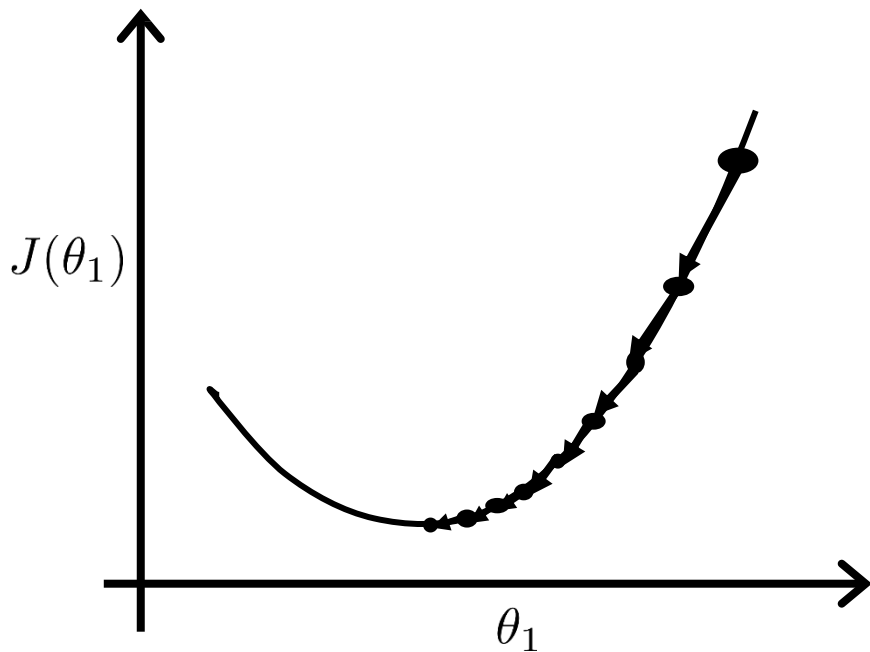


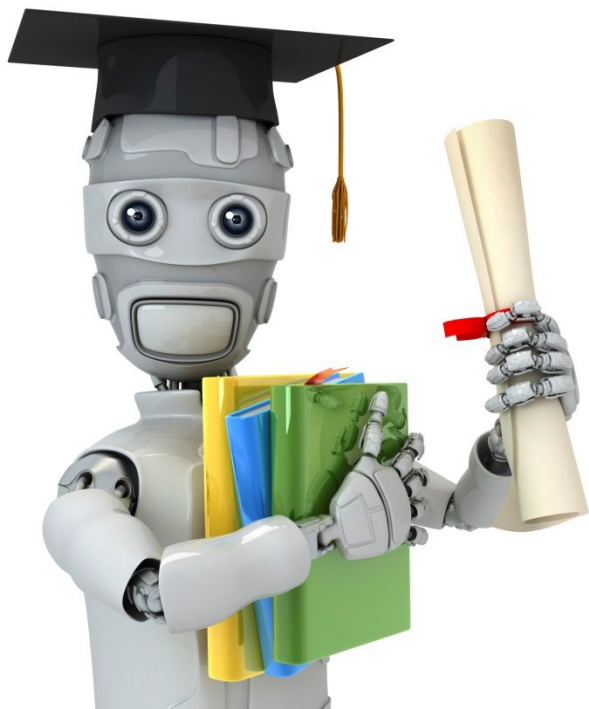
$$\theta_1 := \theta_1 - \alpha \frac{d}{d\theta_1} J(\theta_1)$$

即使学习速率固定，梯度下降也可以收敛到局部最小值。

$$\theta_1 := \theta_1 - \alpha \frac{d}{d\theta_1} J(\theta_1)$$

当我们接近局部最小值时，梯度下降将自动采取更小的步骤。因此，不需要随着时间的推移而减少 α 。





机器学习

单变量线性回归

线性回归中的梯度下降

梯度下降算法

repeat until convergence {
 $\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$
 (for $j = 1$ and $j = 0$)
}

线性回归模型

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$\begin{aligned}
 \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) &= \frac{\partial}{\partial \theta_j} \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 \\
 &= \frac{\partial}{\partial \theta_j} \frac{1}{2m} \sum_{i=1}^m (\theta_0 + \theta_1 x^{(i)} - y^{(i)})^2
 \end{aligned}$$

$$j = 0 : \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$$

$$j = 1 : \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

梯度下降算法

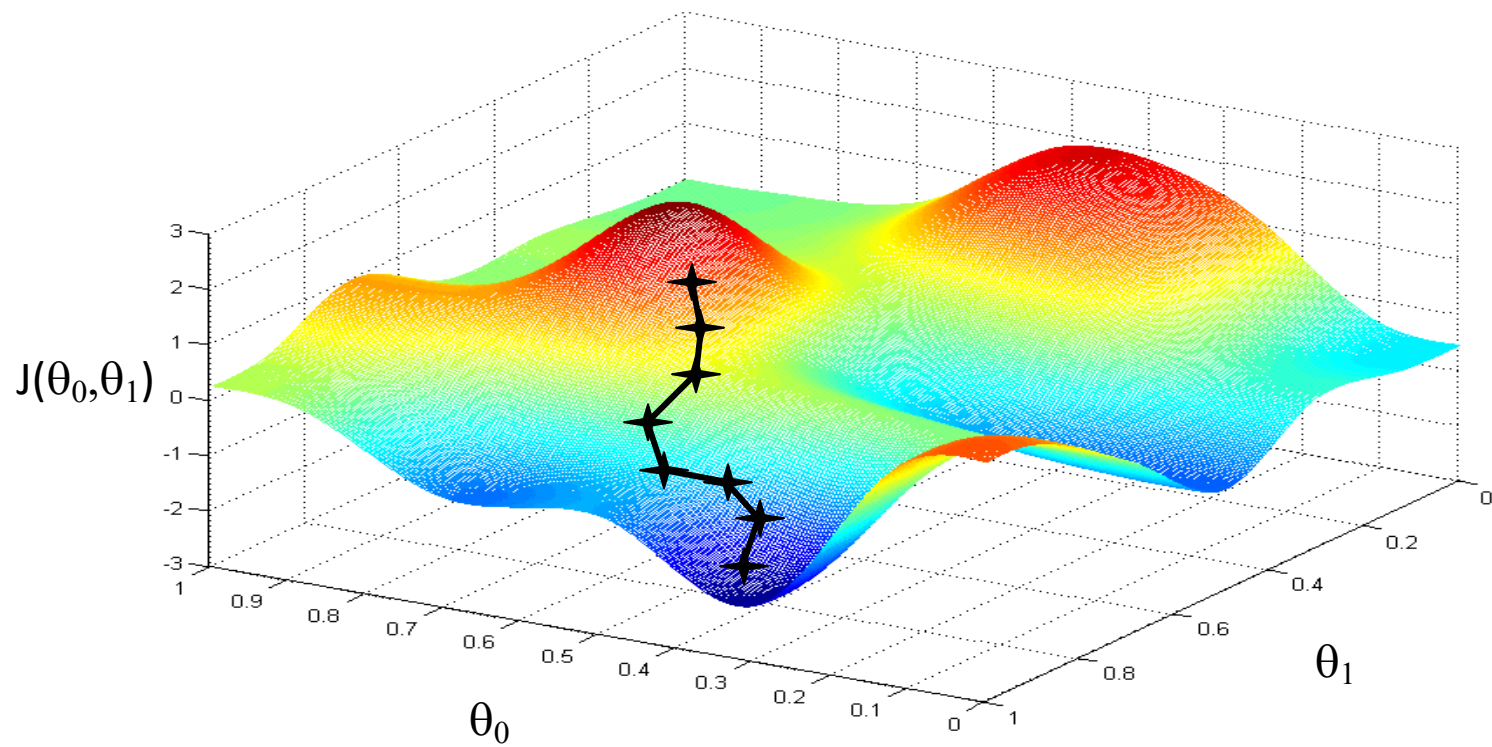
repeat until convergence {

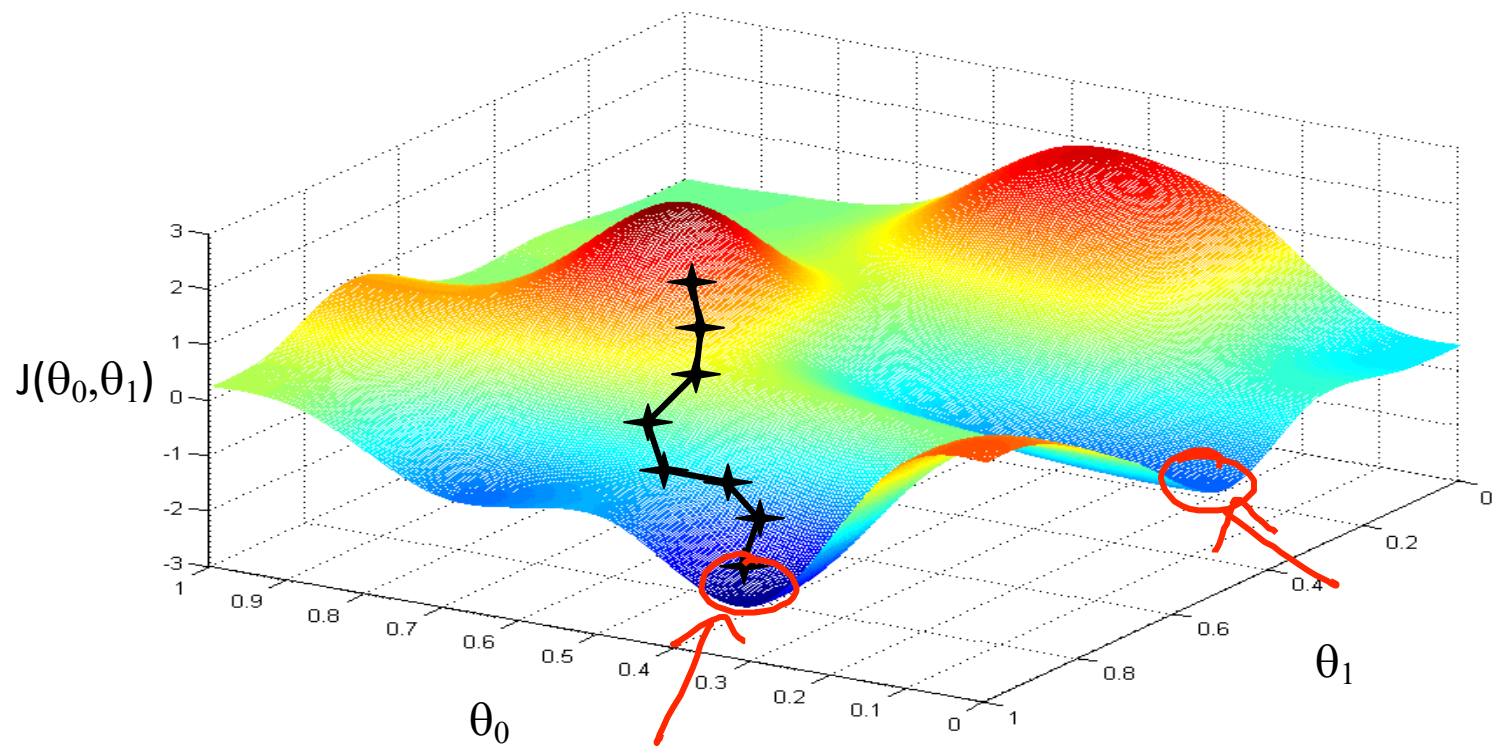
$$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$$

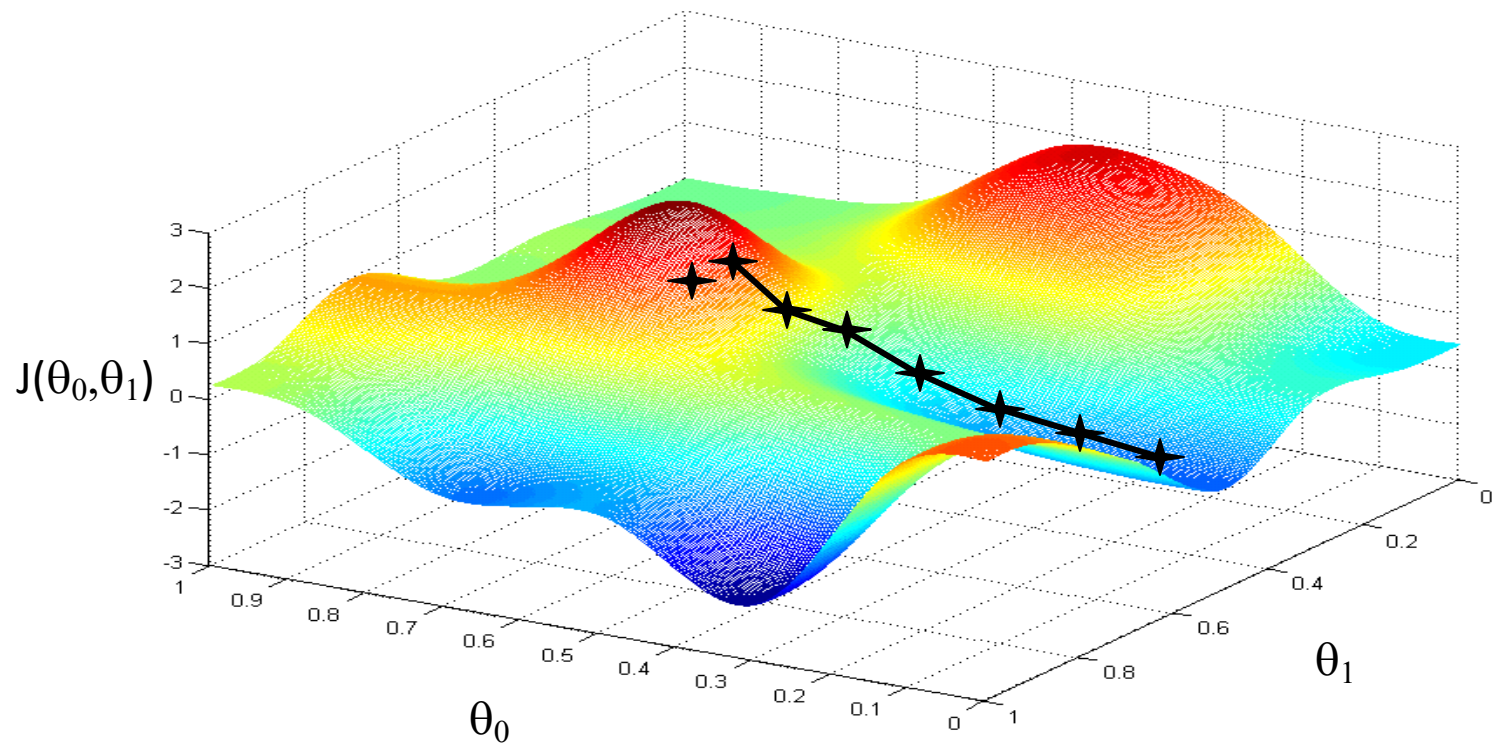
$$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

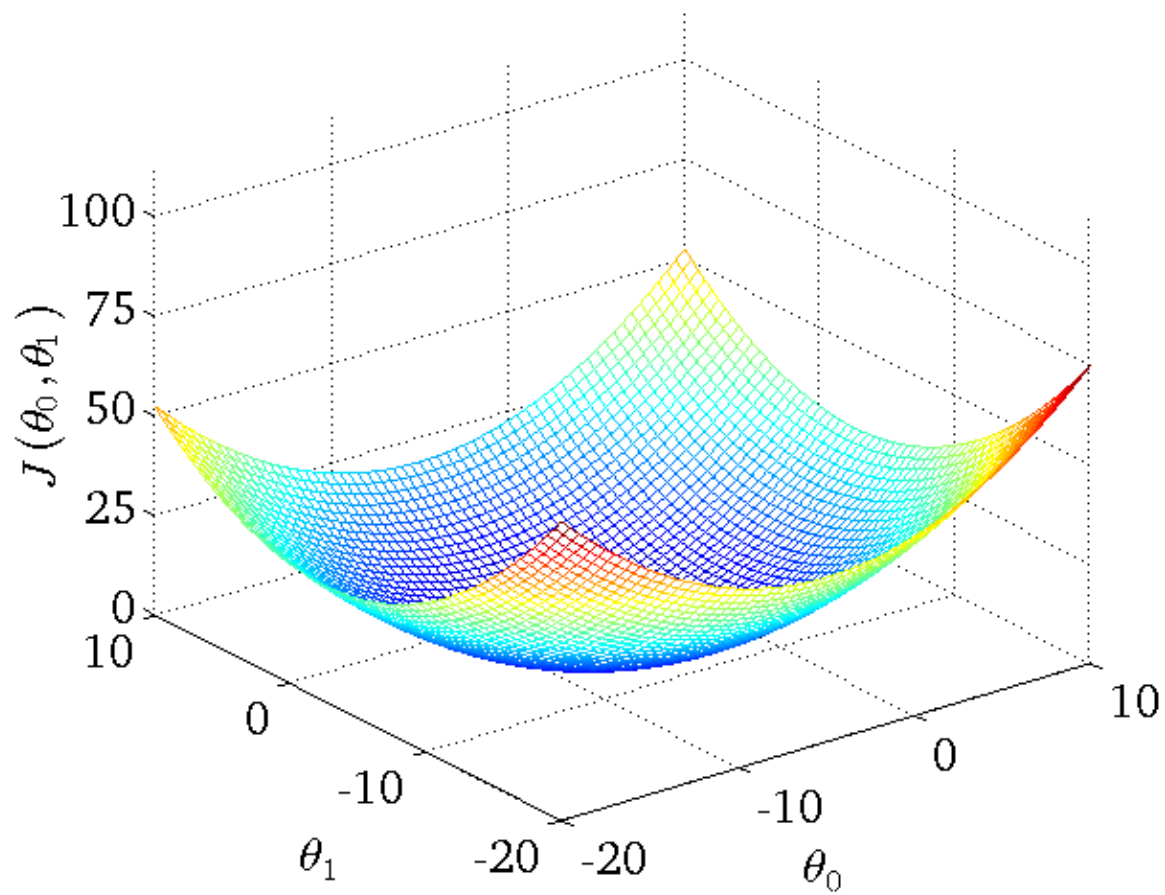
}

同步
更新
 θ_0 和 θ_1



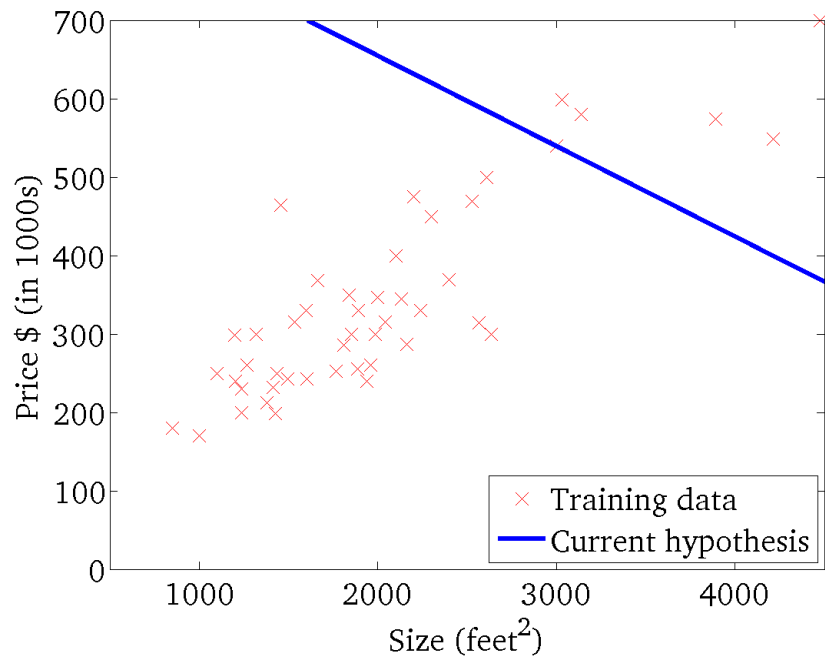






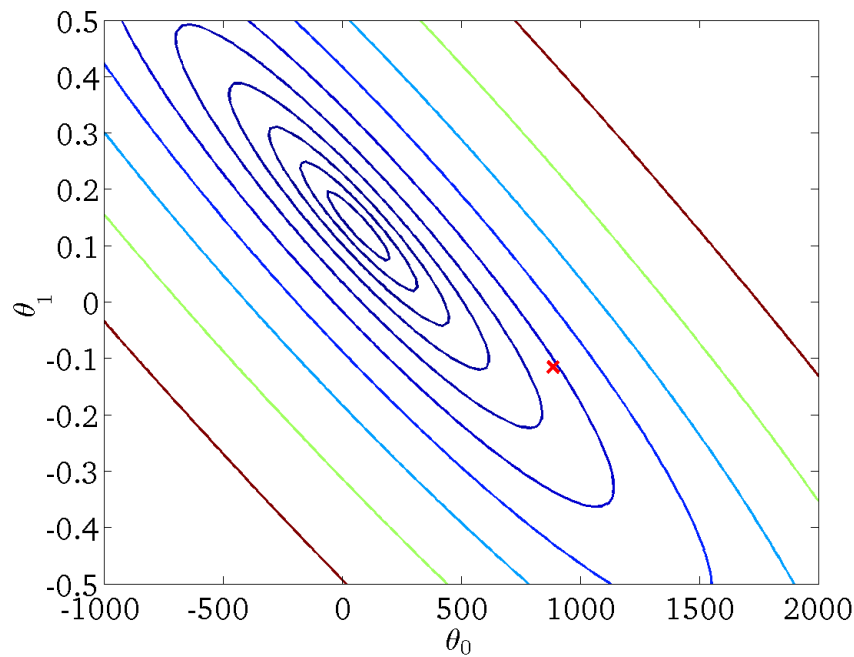
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



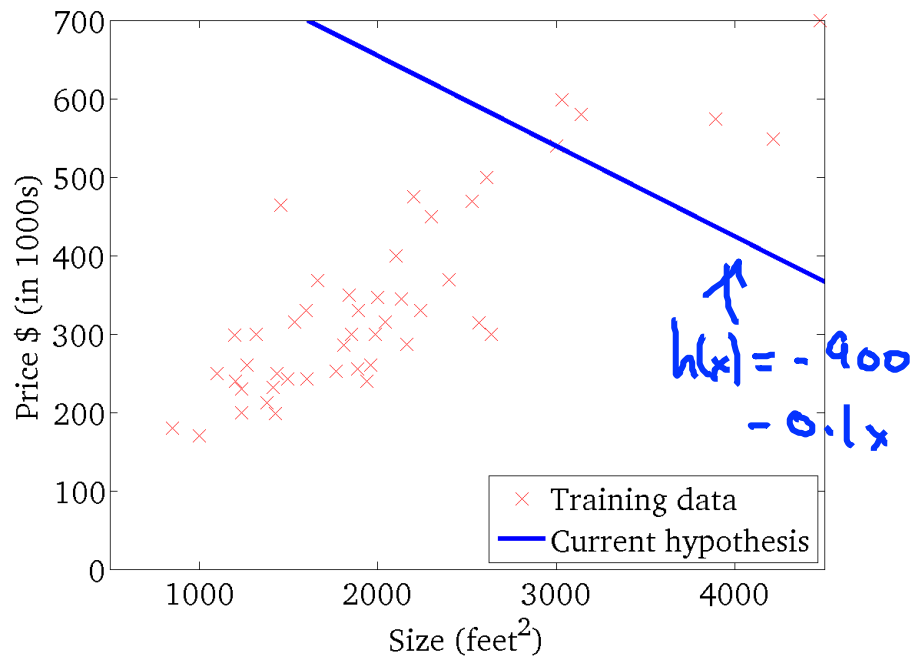
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



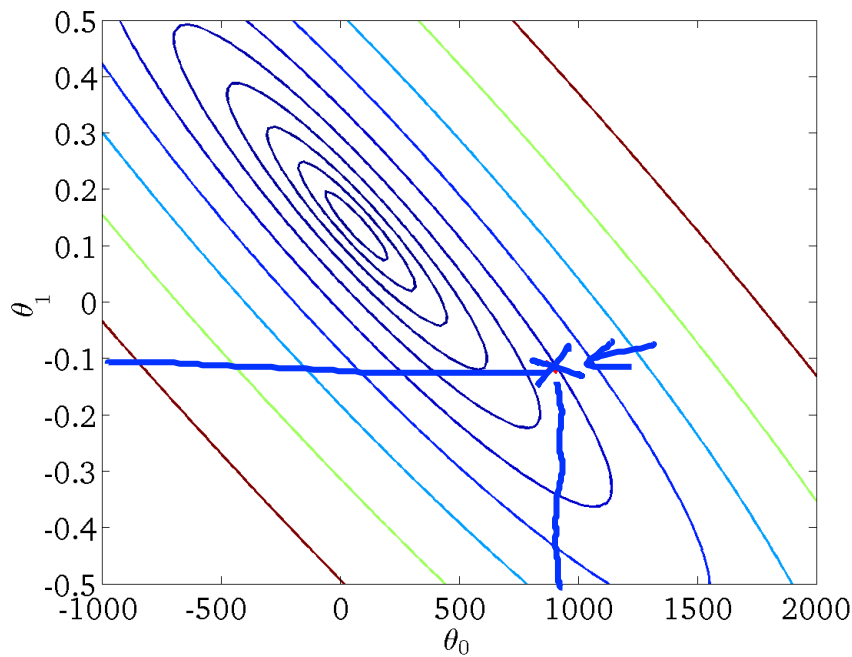
$$\underline{h_{\theta}(x)}$$

(for fixed θ_0, θ_1 , this is a function of x)



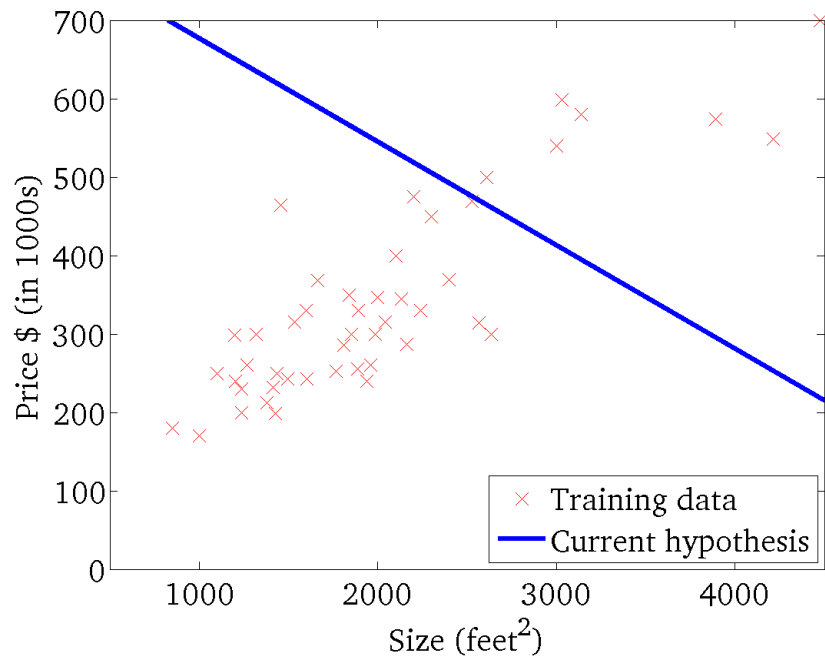
$$\underline{J(\theta_0, \theta_1)}$$

(function of the parameters θ_0, θ_1)



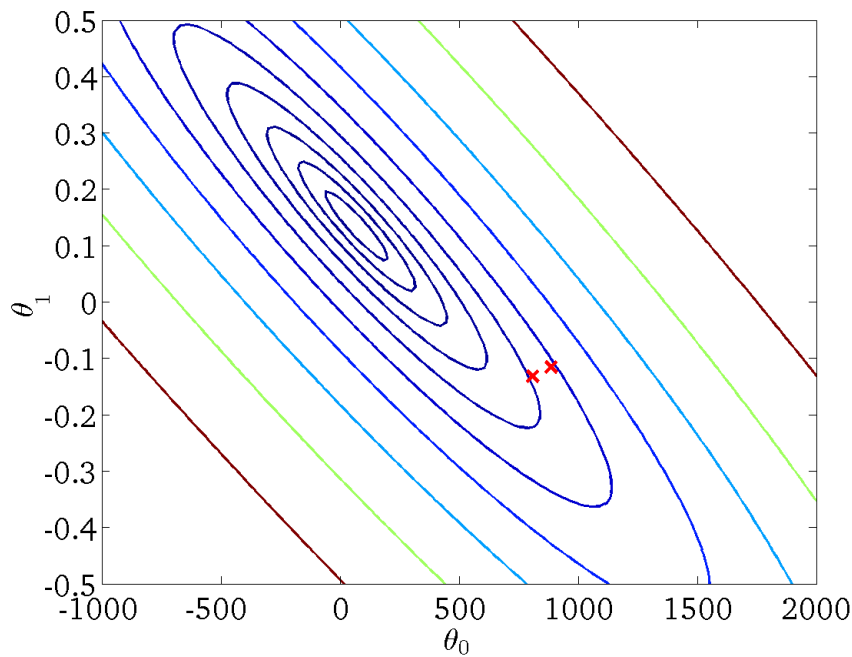
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



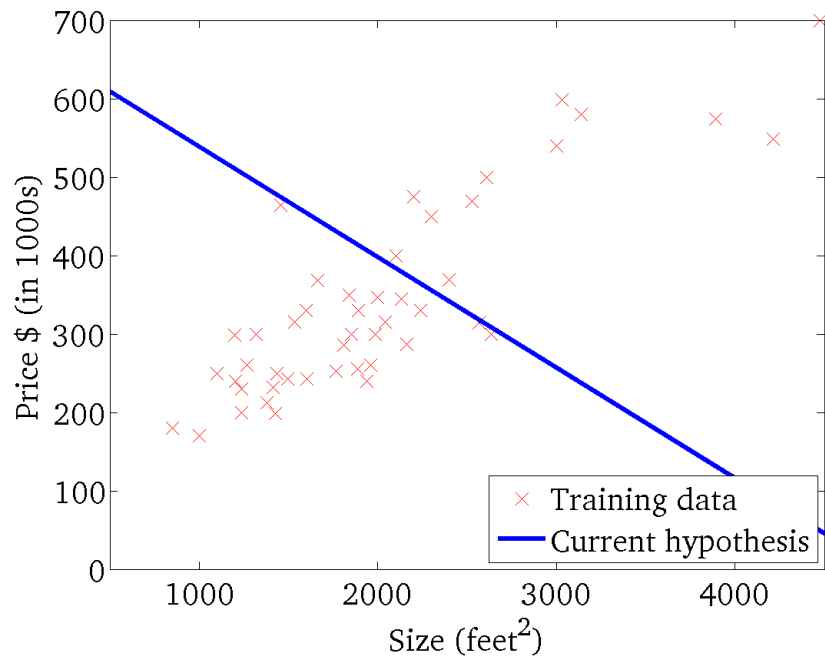
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



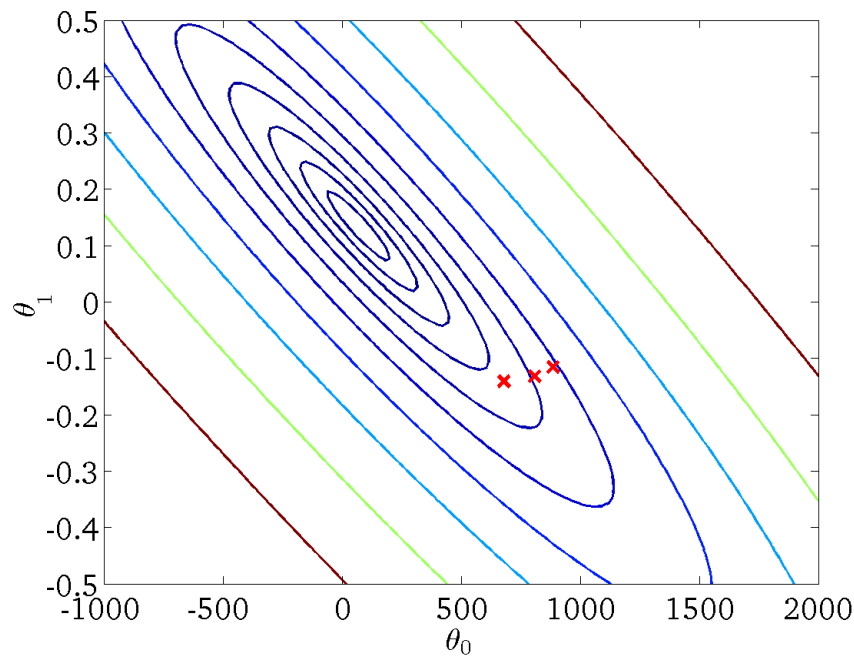
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



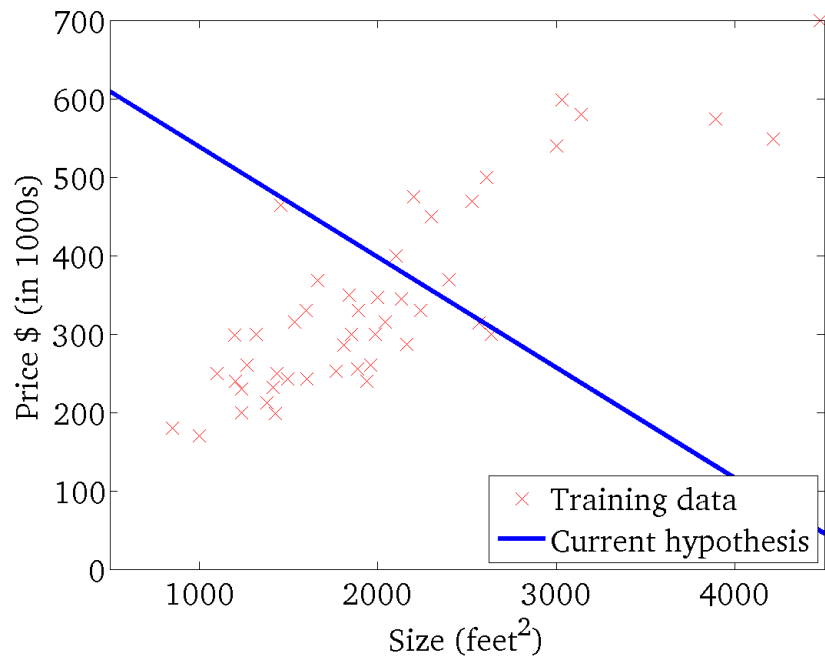
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



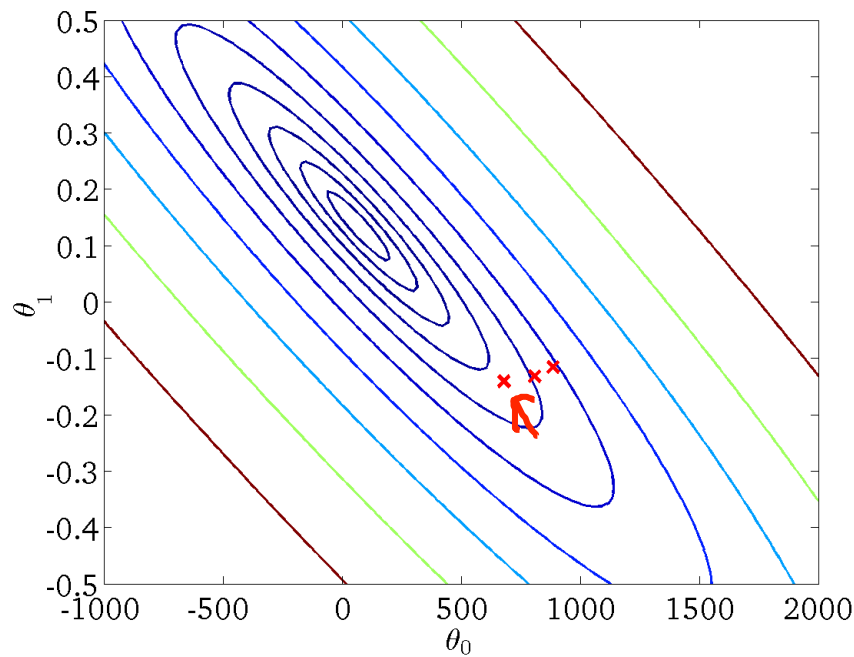
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



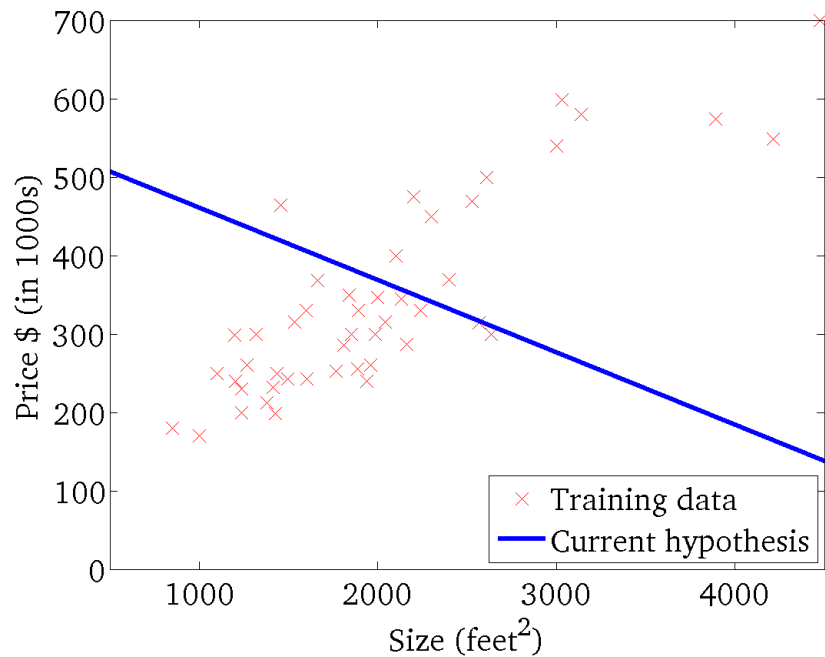
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



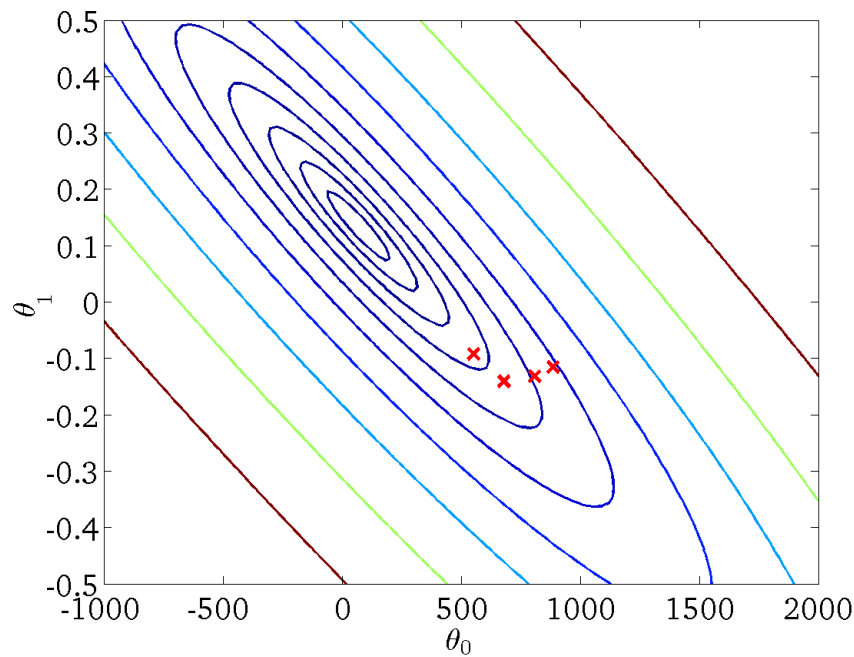
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



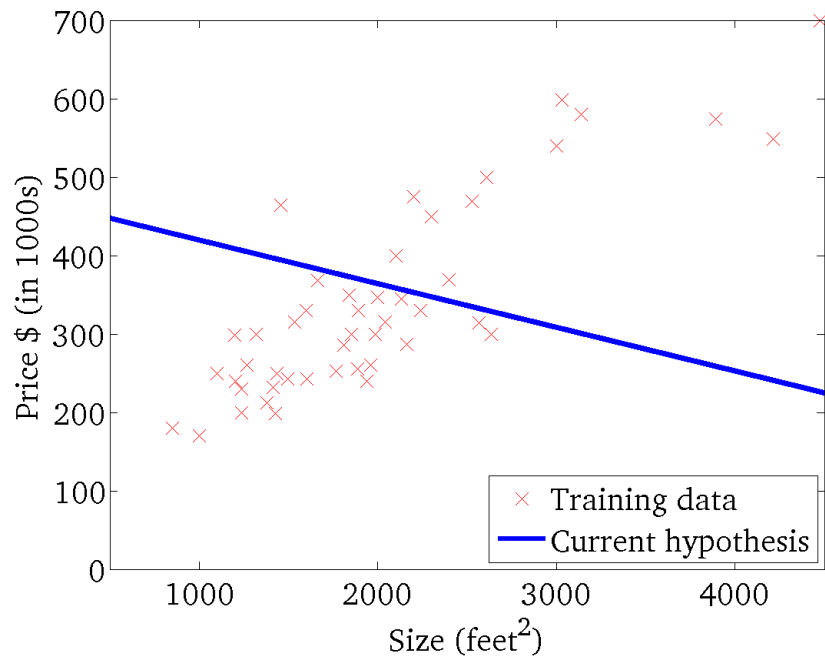
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



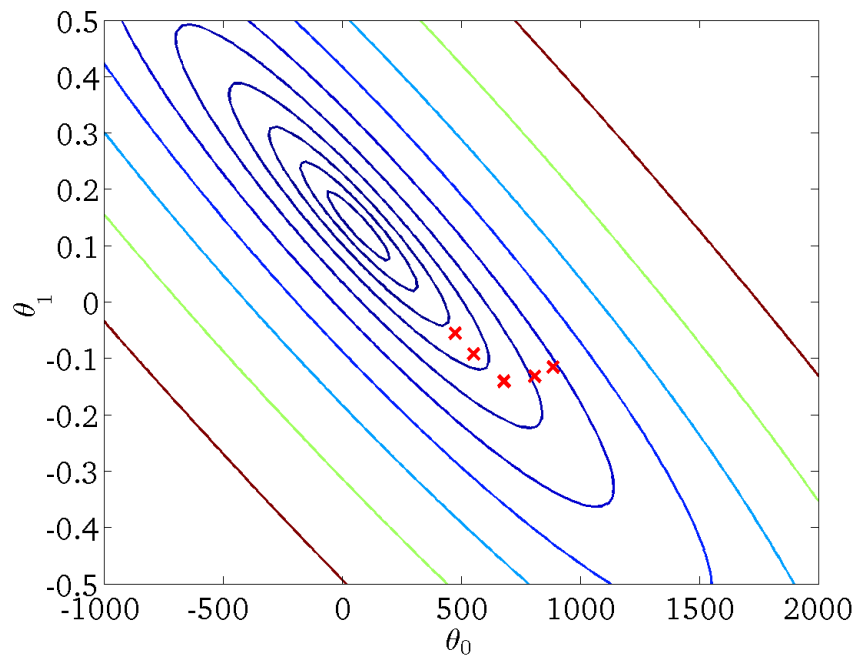
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



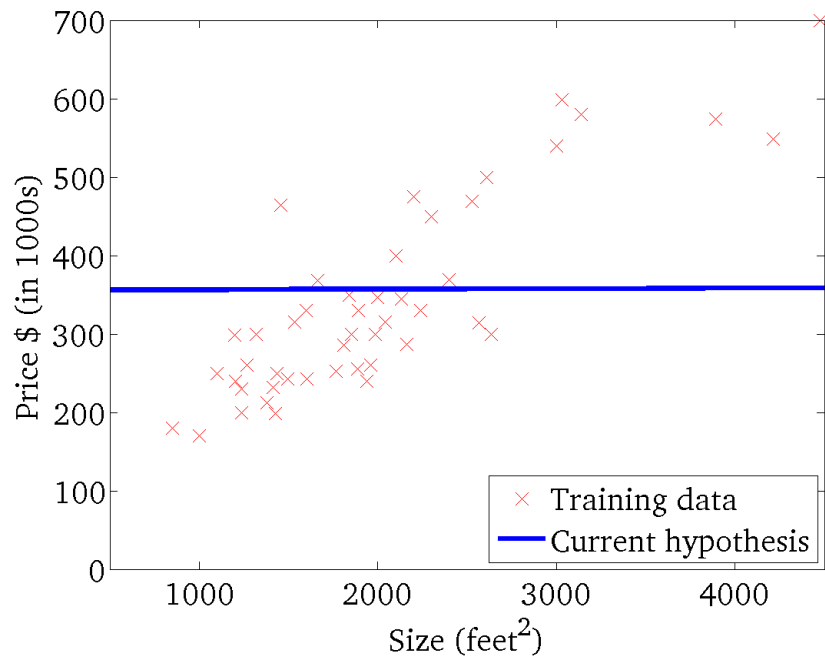
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



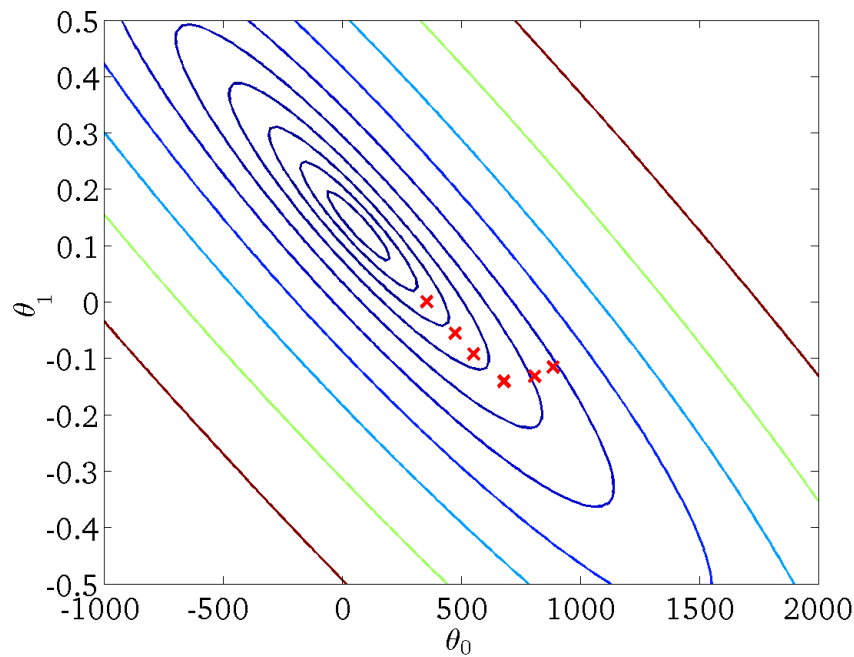
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



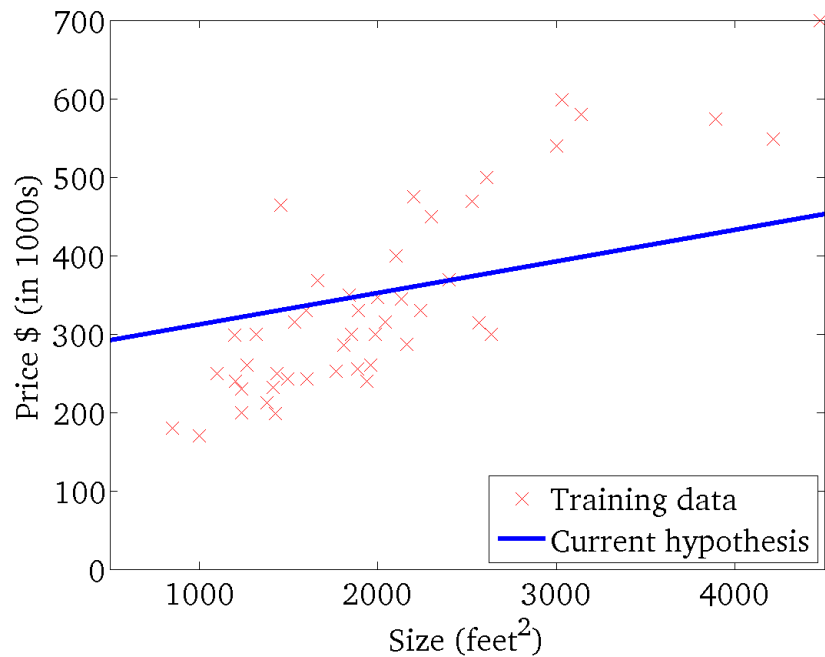
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



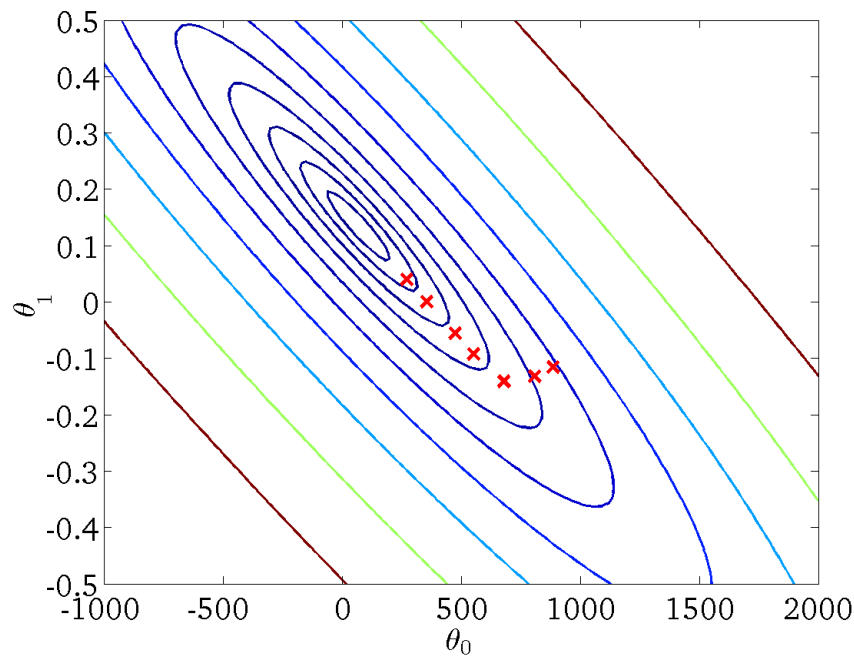
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



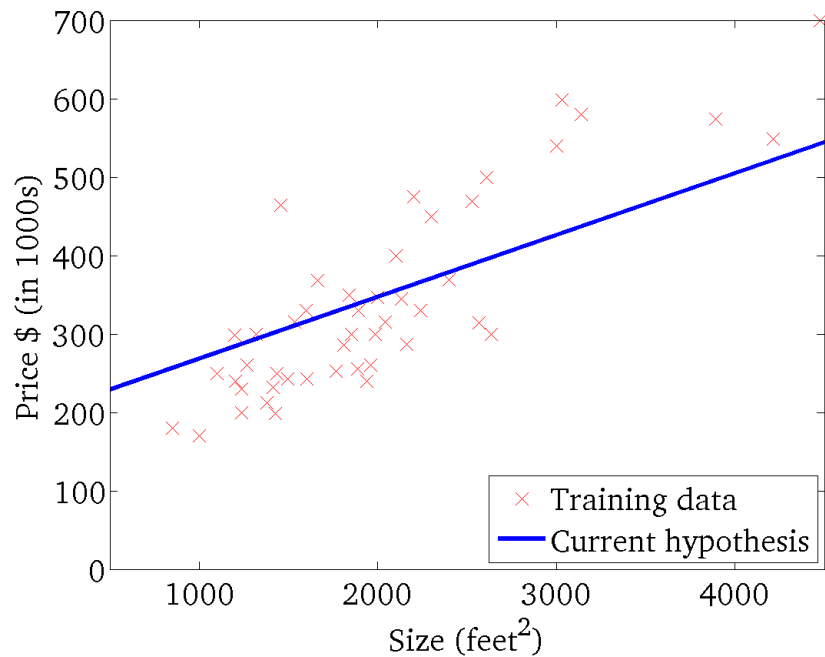
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



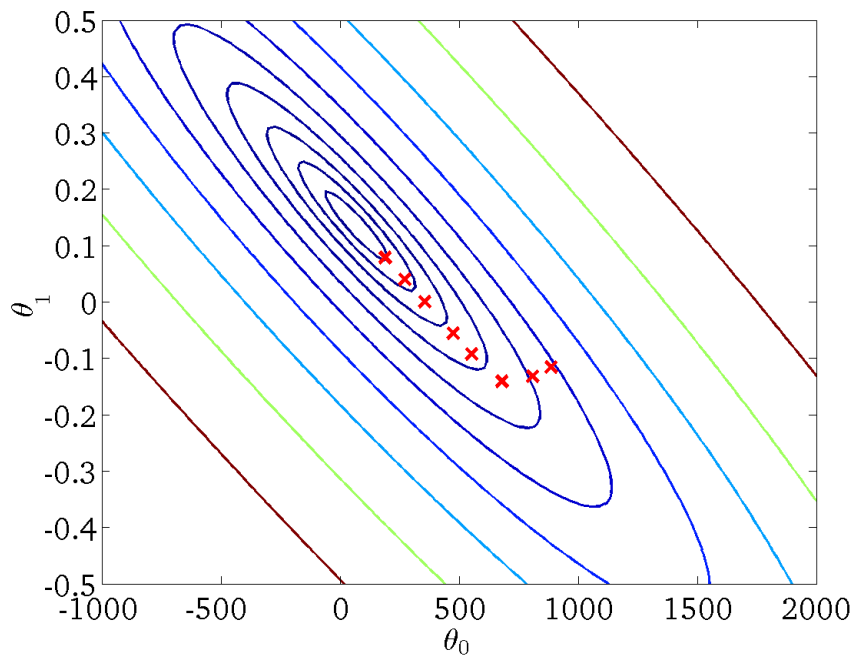
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



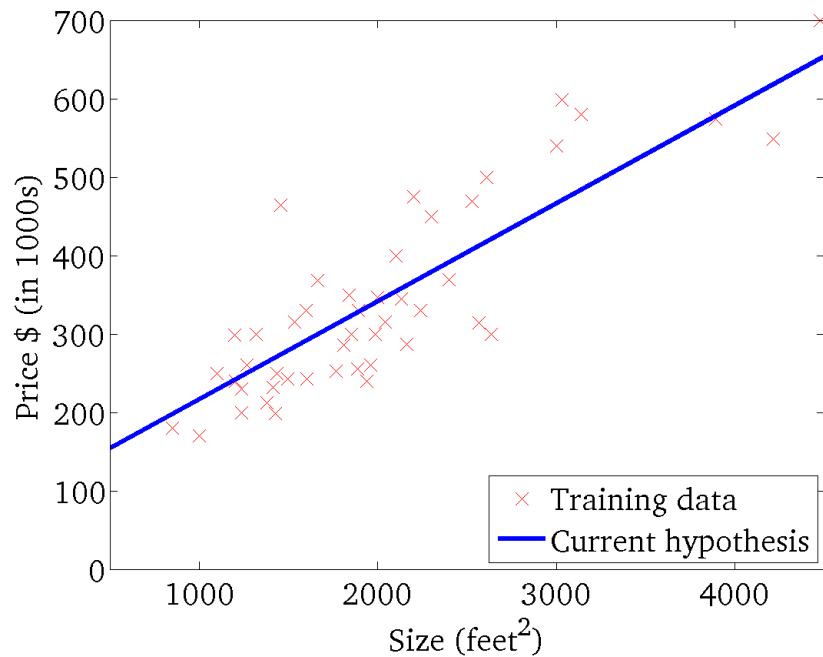
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



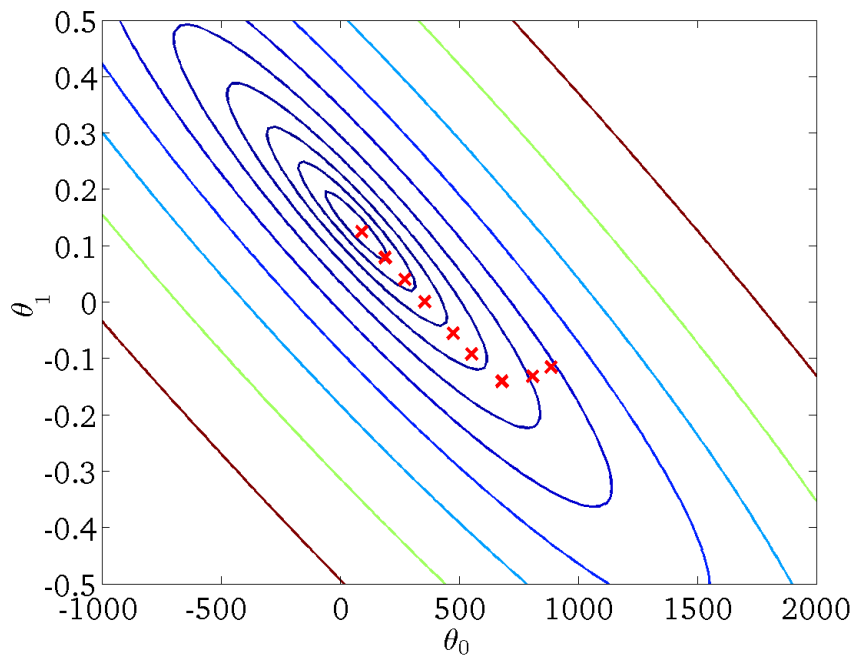
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



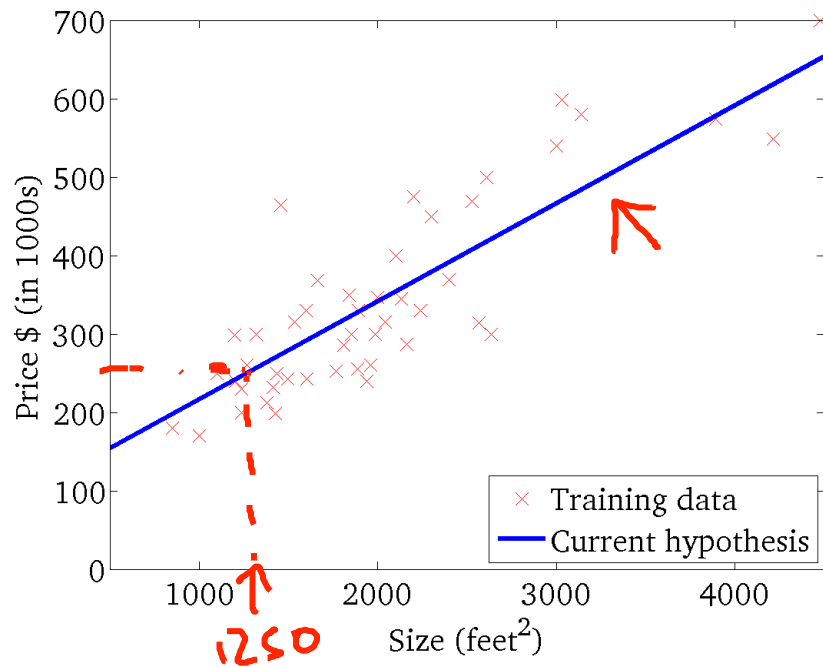
$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



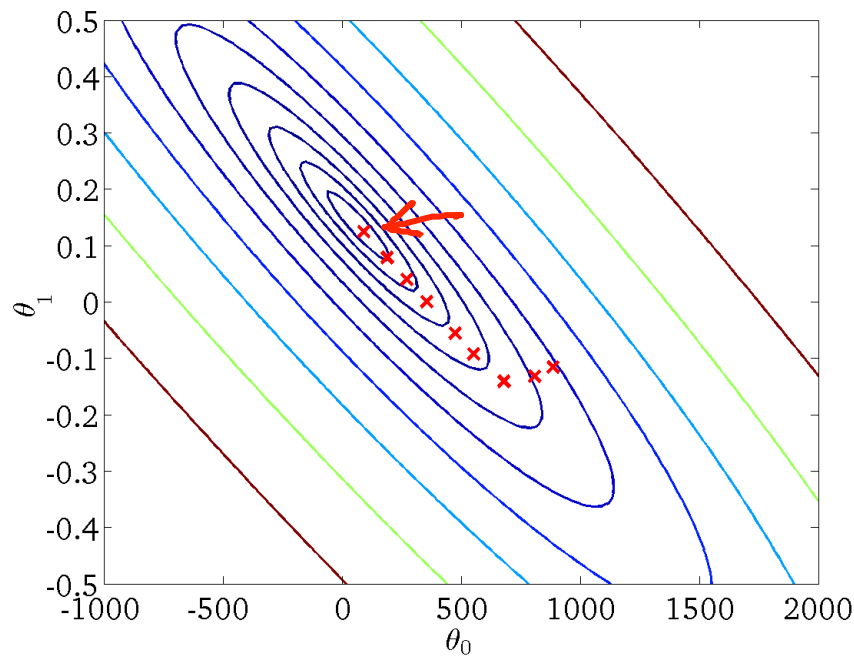
$$h_{\theta}(x)$$

(for fixed θ_0, θ_1 , this is a function of x)



$$J(\theta_0, \theta_1)$$

(function of the parameters θ_0, θ_1)



公式推导小作业2-1:

通过梯度下降法反复迭代找到全局最小值，可否不用迭代方法，直接求出参数 θ 的解析解(闭式解)? 以公式写出原理

假设:
$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

参数:
$$\theta_0, \theta_1$$

代价函数:
$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

目标:
$$\underset{\theta_0, \theta_1}{\text{minimize}} J(\theta_0, \theta_1)$$

提示：根据该提示公式描述求解思路

Matrix notation:

$$\mathbf{y} = \begin{pmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(m)} \end{pmatrix}_{m \times 1} \quad \mathbf{X} = \begin{pmatrix} 1 & x^{(1)} \\ 1 & x^{(2)} \\ \vdots & \vdots \\ 1 & x^{(m)} \end{pmatrix}_{m \times 2} \quad \boldsymbol{\theta} = \begin{pmatrix} \theta_0 \\ \theta_1 \end{pmatrix}_{2 \times 1}$$

The goal is to solve

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta}$$

Note: the system is overdetermined and has no solution since $m > 2$

Example 2.37

Courtesy of the Smithsonian Institution Libraries, Washington, D.C.



Carl Gustav Jacobi (1804–1851) was a German mathematician who made important contributions to many fields of mathematics and physics, including geometry, number theory, analysis, mechanics, and fluid dynamics. Although much of his work was in applied mathematics, Jacobi believed in the importance of doing mathematics for its own sake. A fine teacher, he held positions at the Universities of Berlin and Königsberg and was one of the most famous mathematicians in Europe.

Consider the system

$$7x_1 - x_2 = 5$$

$$3x_1 - 5x_2 = -7$$

Jacobi's method begins with solving the first equation for x_1 and the second equation for x_2 , to obtain

$$\begin{aligned} x_1 &= \frac{5 + x_2}{7} \\ x_2 &= \frac{7 + 3x_1}{5} \end{aligned} \tag{1}$$

We now need an **initial approximation** to the solution. It turns out that it does not matter what this initial approximation is, so we might as well take $x_1 = 0$, $x_2 = 0$. We use these values in Equations (1) to get new values of x_1 and x_2 :

$$x_1 = \frac{5 + 0}{7} = \frac{5}{7} \approx 0.714$$

$$x_2 = \frac{7 + 3 \cdot 0}{5} = \frac{7}{5} = 1.400$$

Now we substitute these values into (1) to get

$$x_1 = \frac{5 + 1.4}{7} \approx 0.914$$

$$x_2 = \frac{7 + 3 \cdot \frac{5}{7}}{5} \approx 1.829$$

(written to three decimal places). We repeat this process (using the old values of x_2 and x_1 to get the new values of x_1 and x_2), producing the sequence of approximations given in Table 2.12.

Table 2.12

n	0	1	2	3	4	5	6
x_1	0	0.714	0.914	0.976	0.993	0.998	0.999
x_2	0	1.400	1.829	1.949	1.985	1.996	1.999

The successive vectors $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ are called *iterates*, so, for example, when $n = 4$, the fourth iterate is $\begin{bmatrix} 0.993 \\ 1.985 \end{bmatrix}$. We can see that the iterates in this example are

→ approaching $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, which is the exact solution of the given system. (Check this.)

We say in this case that Jacobi's method *converges*.



Jacobi's method calculates the successive iterates in a two-variable system according to the crisscross pattern shown in Table 2.13.

Table 2.13

n	0	1	2	3
x_1		→	→	→
x_2		→	→	→

$$1 \quad 7x_1 - x_2 = 5$$

$$3x_1 - 5x_2 = -7$$

$$x_1 = \frac{5 + x_2}{7}$$

2

$$x_2 = \frac{7 + 3x_1}{5}$$

$$x_1 = \frac{5 + 0}{7} = \frac{5}{7} \approx 0.714$$

3

$$x_2 = \frac{7 + 3 \cdot 0}{5} = \frac{7}{5} = 1.400$$

4

$$x_2 = \frac{7 + 3 \cdot \frac{5}{7}}{5} \approx 1.829$$

The Gauss-Seidel method is named after C. F. Gauss and [Philipp Ludwig von Seidel \(1821–1896\)](#). Seidel worked in analysis, probability theory, astronomy, and optics. Unfortunately, he suffered from eye problems and retired at a young age. The paper in which he described the method now known as Gauss-Seidel was published in 1874. Gauss, it seems, was unaware of the method!

Table 2.14

n	0	1	2	3	4	5
x_1	0	0.714	0.976	0.998	1.000	1.000
x_2	0	1.829	1.985	1.999	2.000	2.000

Table 2.12

n	0	1	2	3	4	5	6
x_1	0	0.714	0.914	0.976	0.993	0.998	0.999
x_2	0	1.400	1.829	1.949	1.985	1.996	1.999

Table 2.15

n	0	1	2	3
x_1				
x_2				

The diagram illustrates a sequence of points (x_1, x_2) for $n=0, 1, 2, 3$. The points are connected by arrows, showing a zig-zag pattern. The arrows for x_1 point right, and the arrows for x_2 point down. The sequence starts at $(0,0)$ and moves towards $(3,3)$.

$$x_1 = \frac{5 + x_2}{7}$$

$$x_2 = \frac{7 + 3x_1}{5}$$

Table 2.14

n	0	1	2	3
x_1	0	0.714	0.976	0.998
x_2	0	1.829	1.985	1.999

Table 2.15

n	0	1	2	3
x_1				
x_2				

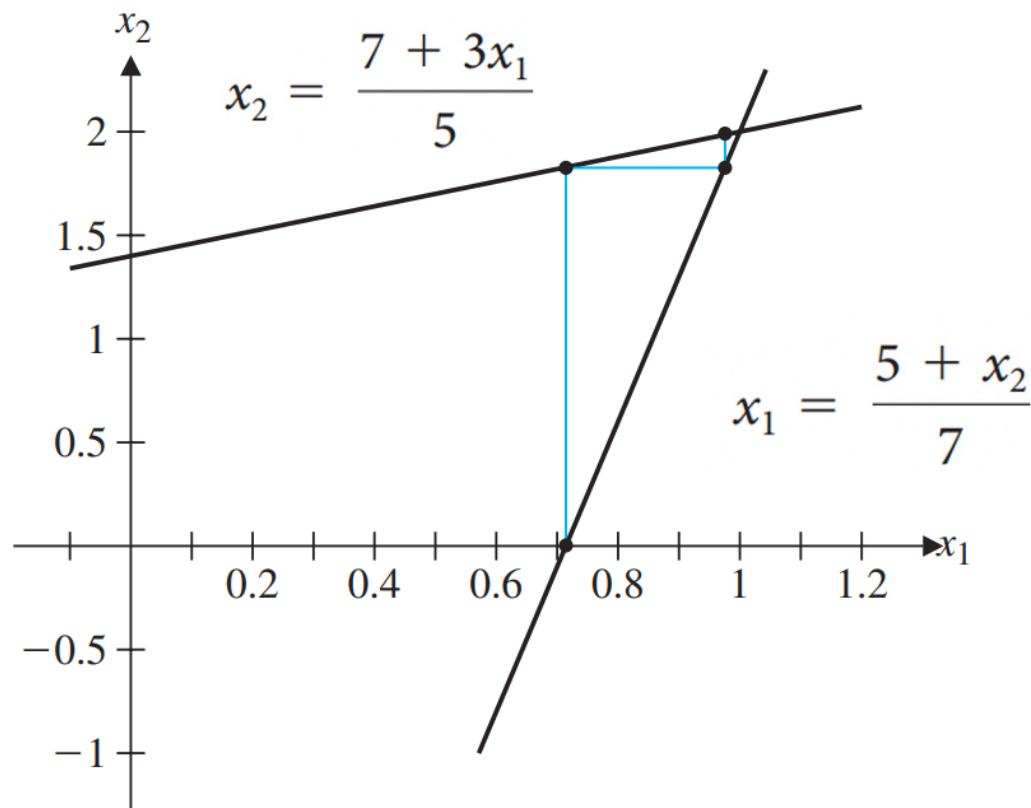


Figure 2.27

Converging iterates